

Metacognition_change

Loading data

Day 1

Day 3

Demographics

Adding errors

Separating overestimating kids

Correlation matrix

```
##          Day1_Acc Day1_Postdiction Day3_Acc Day3_Postdiction  MA  MV
## Day1_Acc          1.00          0.32      0.64          0.46 -0.12 0.15
## Day1_Postdiction  0.32          1.00      0.27          0.40 -0.12 0.22
## Day3_Acc          0.64          0.27      1.00          0.58 -0.09 0.09
## Day3_Postdiction  0.46          0.40      0.58          1.00 -0.11 0.20
## MA               -0.12         -0.12     -0.09          -0.11  1.00 0.03
## MV               0.15          0.22      0.09          0.20  0.03 1.00
##
## n
##          Day1_Acc Day1_Postdiction Day3_Acc Day3_Postdiction  MA  MV
## Day1_Acc          441          441      441          441 437 434
## Day1_Postdiction  441          441      441          441 437 434
## Day3_Acc          441          441      441          441 437 434
## Day3_Postdiction  441          441      441          441 437 434
## MA               437          437      437          437 437 434
## MV               434          434      434          434 434 434
##
## P
##          Day1_Acc Day1_Postdiction Day3_Acc Day3_Postdiction  MA
## Day1_Acc          0.0000          0.0000  0.0000          0.0136
## Day1_Postdiction 0.0000          0.0000  0.0000          0.0154
## Day3_Acc          0.0000  0.0000          0.0000          0.0539
## Day3_Postdiction 0.0000  0.0000          0.0000          0.0281
## MA               0.0136  0.0154          0.0539  0.0281
## MV               0.0015  0.0000          0.0504  0.0000          0.5387
##          MV
## Day1_Acc          0.0015
## Day1_Postdiction 0.0000
## Day3_Acc          0.0504
```

##

##

##

##

##

##

##

##

##

##

##

##

||

##

##

##

##

##

##

##

##

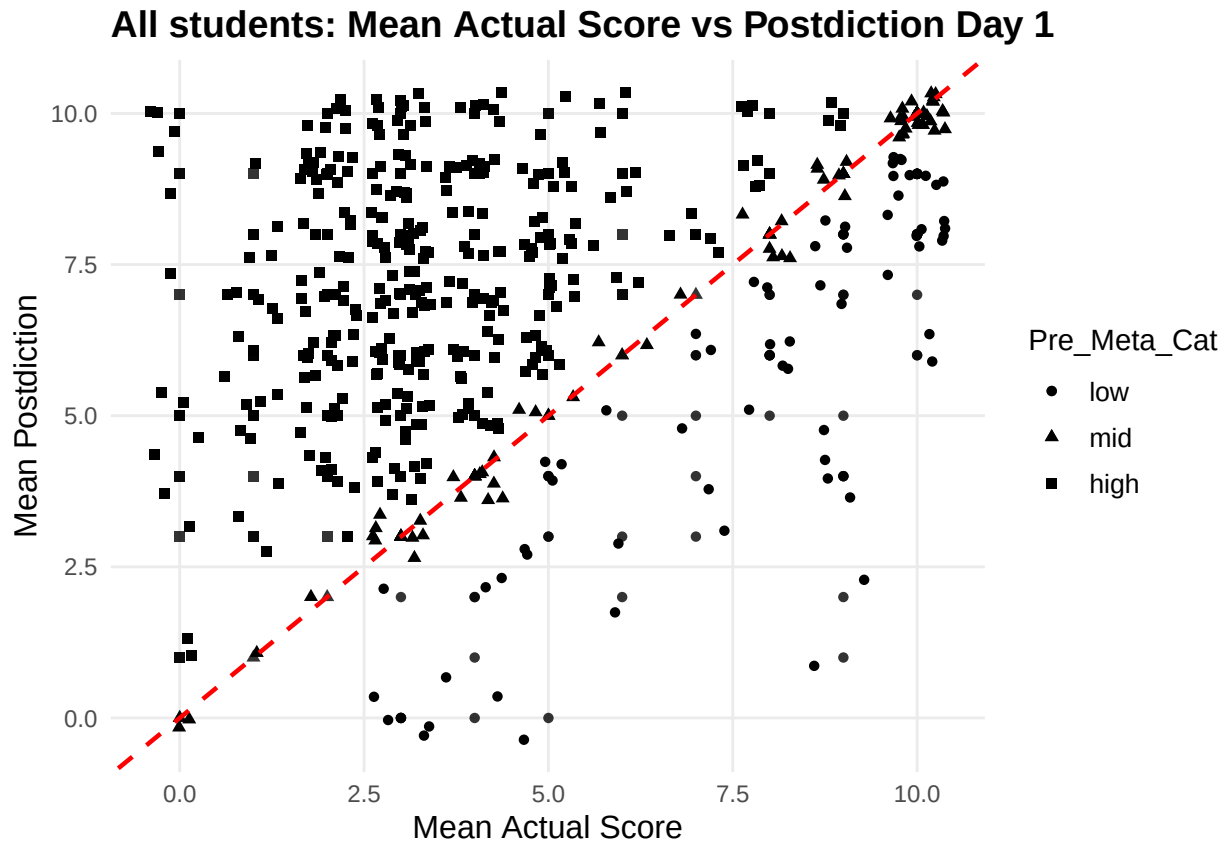
##

##

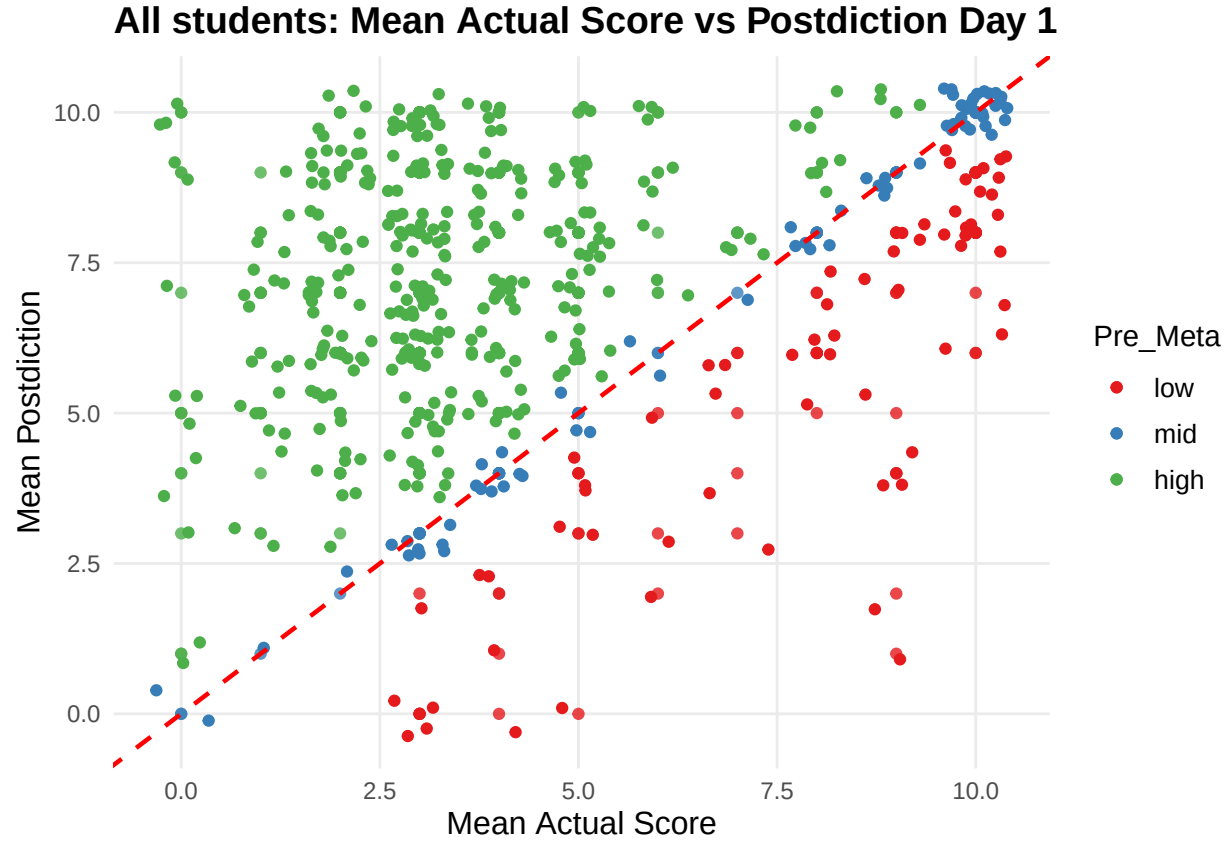
Visualizations - all kids

Ac score vs postdiction plots for all kids

```
## Ignoring unknown labels:  
## * note : "Red line represents perfect alignment between actual score and  
## postdictions, dots above the red line represent overestimating responses and  
## below the red line - underestimating responses"
```



```
## Ignoring unknown labels:  
## * note : "Red line represents perfect alignment between actual score and  
## postdictions, dots above the red line represent overestimating responses and  
## below the red line - underestimating responses"
```



Tables by initial overestimation group

```
##
##          NSNC NSCC NSIC CSNC CSCC CSIC ISNC ISCC ISIC
## low       13  17  13  10   0   0  10   0   0
## mid       12  15  11  12   0   0  14   0   0
## high      64  58  75  51   0   0  66   0   0
```

```
##
## Attaching package: 'kableExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##      group_rows
```

Table 1: Descriptive Statistics by Pre-test Metacognition Category

Pre-Meta Category	N	MA M(SD)	MV M(SD)	Day1 Accuracy M(SD)	Day1 Postdiction M(SD)	Day1 Metacognition M(SD)
low	63	1.45 (0.41)	2.11 (0.66)	7.75 (2.38)	5.41 (2.94)	-2.33 (1.53)
mid	64	1.39 (0.38)	2.08 (0.73)	7.11 (3.18)	7.11 (3.18)	0 (0)
high	314	1.43 (0.45)	2.08 (0.62)	3.25 (1.68)	7.27 (1.88)	4.02 (2.06)
Overall	441	1.43 (0.44)	2.08 (0.64)	4.45 (2.8)	6.98 (2.36)	2.53 (3.04)

```
##
##      low mid high Sum
##  0      24  30 140 194
##  1      34  30 146 210
## <NA>    5   4  28  37
## Sum    63  64 314 441
```

```
##
##              low mid high Sum
## American Indian    0   0   1   1
## Asian              47  35  95 177
## Black              1   2  10  13
## Hispanic Latino    2   7  64  73
## Multiple Races     1   2   8  11
## White              7  14 108 129
## <NA>                5   4  28  37
## Sum                63  64 314 441
```

Tables

Day 1 to day 3 table

All kids

```
## # A tibble: 5 x 4
##   Condition Day1_Metacognition Day3_Metacognition Acc_Change
##   <fct>          <dbl>          <dbl>          <dbl>
## 1 NSNC              2.8              1.1              0.8
## 2 NSCC              2                1.6              0.7
## 3 NSIC              2.8              1.1              1.2
## 4 CSNC              2.3              0.7              1.5
## 5 ISNC              2.6              1.2              1
```

Overestimating students

```
## # A tibble: 5 x 4
##   Condition Day1_Metacognition Day3_Metacognition Acc_Change
##   <fct>          <dbl>          <dbl>          <dbl>
## 1 NSNC              4.3              1.7              1
## 2 NSCC              3.7              2.2              1.3
## 3 NSIC              4.2              1.3              1.8
## 4 CSNC              3.8              0.9              2.1
## 5 ISNC              4                1.8              1.2
```

Change table for initially overestimating kids

```
## # A tibble: 5 x 8
##   Condition      n mean_Acc_change mean_Postd mean_Meta_change sd_Acc_change
##   <fct>    <int>          <dbl>          <dbl>          <dbl>          <dbl>
## 1 NSNC      128          0.953          -1.62          -1.70          0.234
## 2 NSCC      116          1.33           -0.207          -0.672          0.227
```

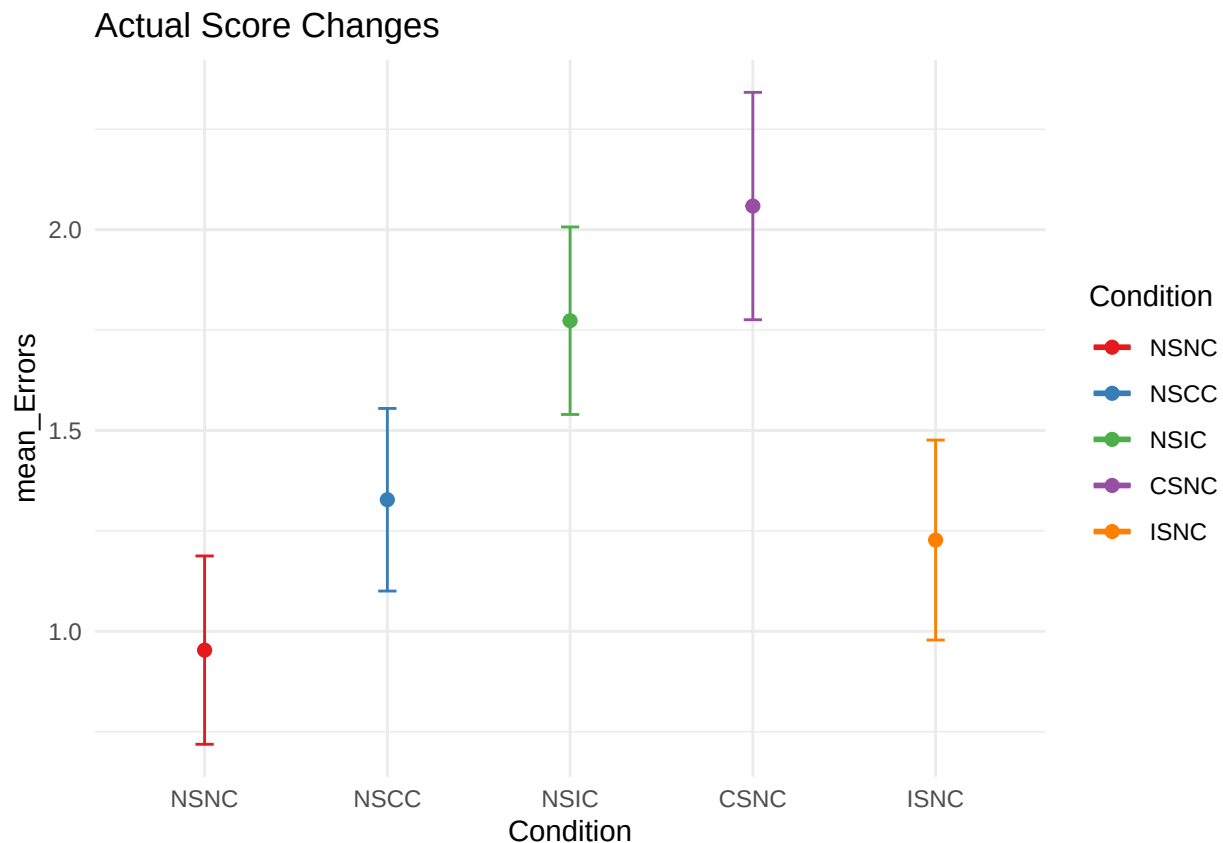
Condition	n	mean_Acc_change	mean_Postd	mean_Meta_change	sd_Acc_change	sd_Postd_change	sd_Meta_change
NSNC	128	0.953125	-1.6250000	-1.7031250	0.2344734	0.2584688	0.2701886
NSCC	116	1.327586	-0.2068966	-0.6724138	0.2273555	0.1814143	0.2123531
NSIC	150	1.773333	-1.0400000	-1.8800000	0.2334733	0.2142319	0.2119305
CSNC	102	2.058823	-0.8235294	-1.4705882	0.2830978	0.2345650	0.2367457
ISNC	132	1.227273	-0.9545455	-1.4242424	0.2488909	0.2688345	0.2643349

```
## 3 NSIC      150      1.77      -1.04      -1.88      0.233
## 4 CSNC      102      2.06      -0.824     -1.47      0.283
## 5 ISNC      132      1.23      -0.955     -1.42      0.249
##   sd_Postd_change sd_Meta_change
##               <dbl>         <dbl>
## 1              0.258             0.270
## 2              0.181             0.212
## 3              0.214             0.212
## 4              0.235             0.237
## 5              0.269             0.264
```

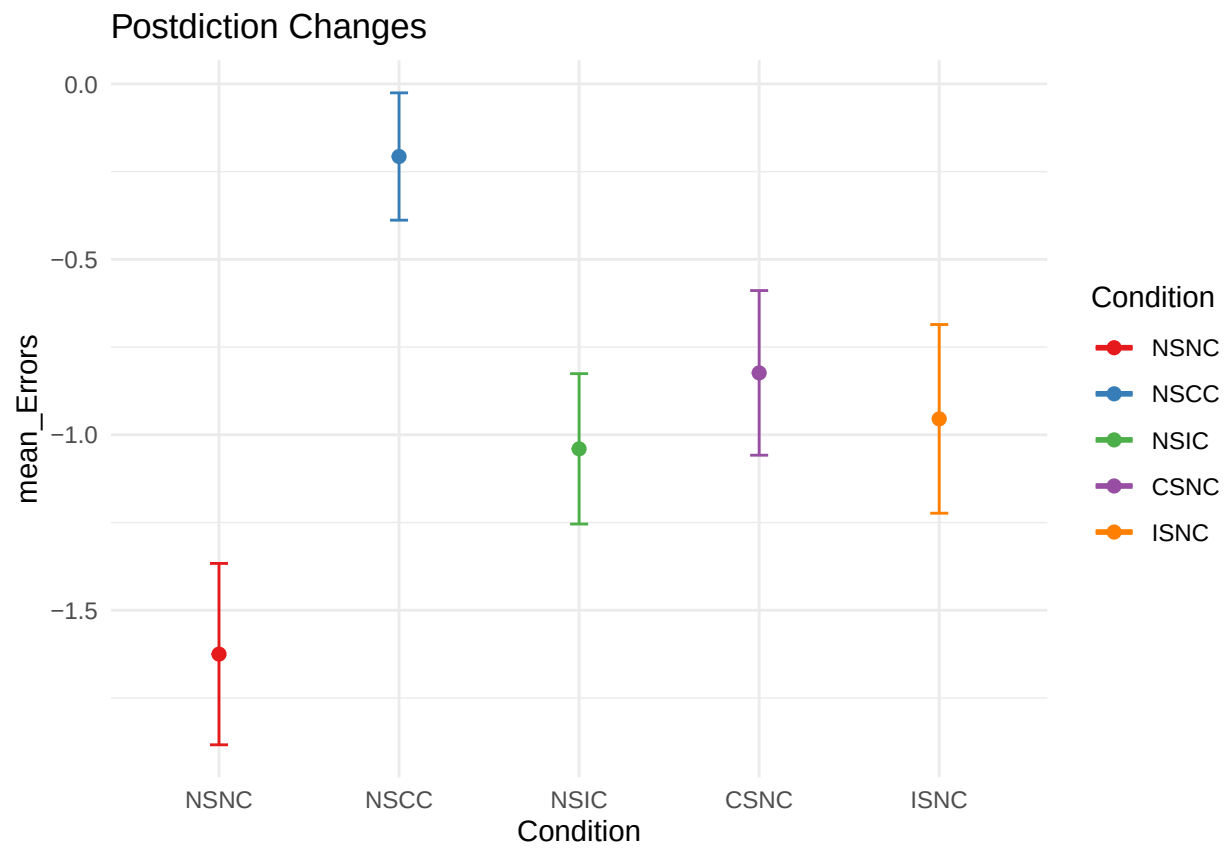
Visualizations - overestimating kids

Bar charts

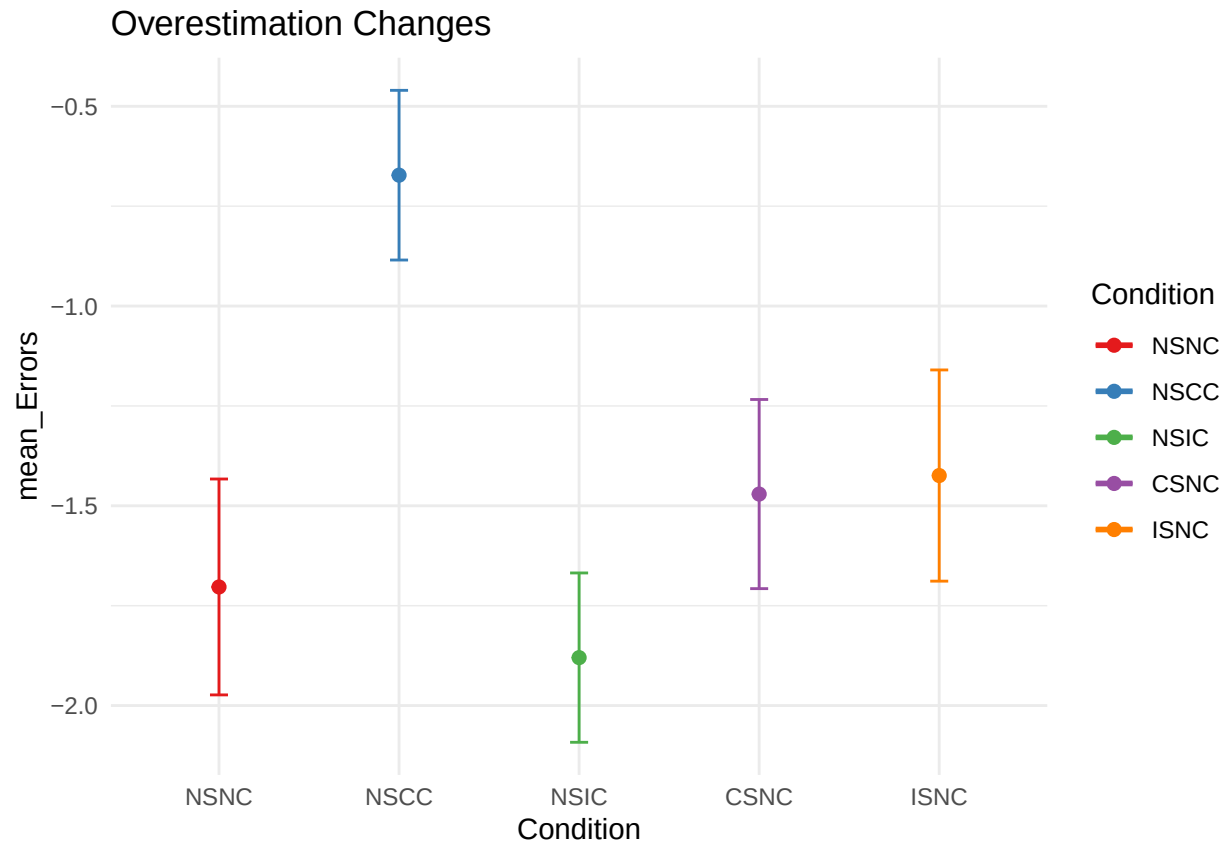
```
## 'geom_line()': Each group consists of only one observation.
## i Do you need to adjust the group aesthetic?
```



```
## 'geom_line()': Each group consists of only one observation.  
## i Do you need to adjust the group aesthetic?
```

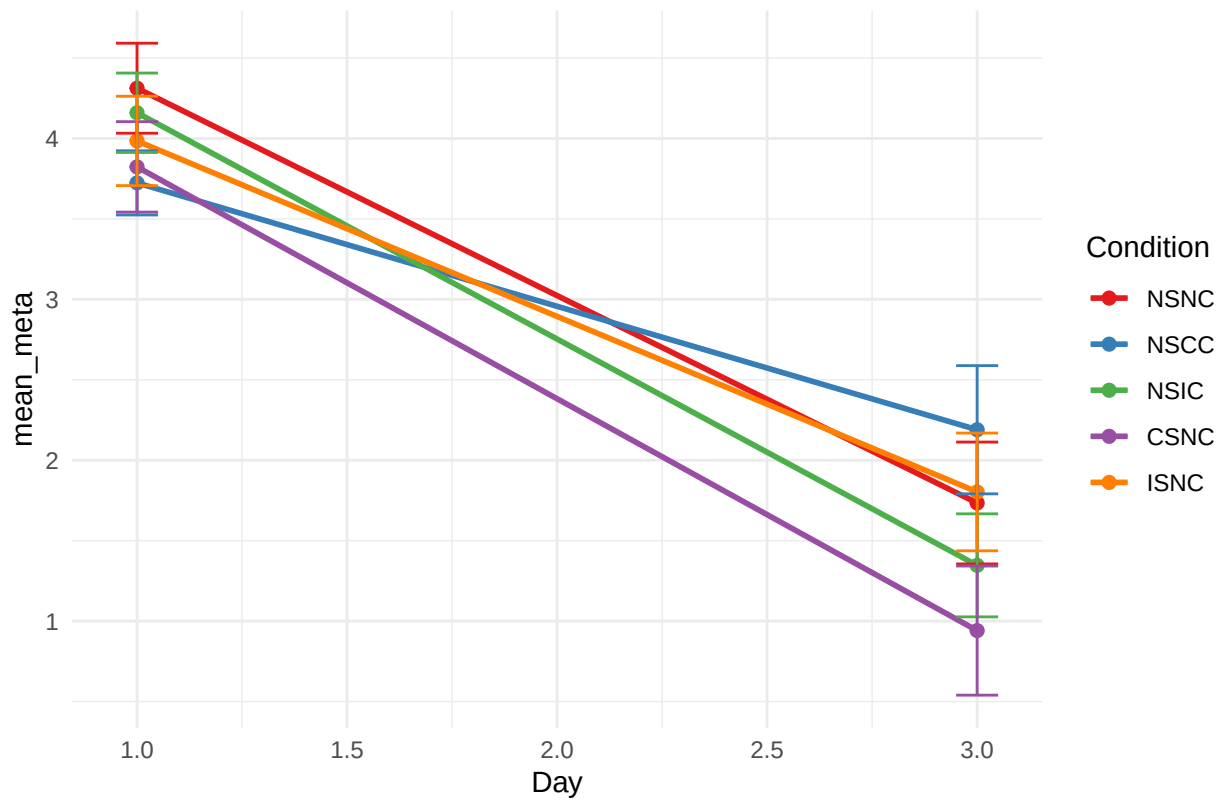


```
## 'geom_line()': Each group consists of only one observation.  
## i Do you need to adjust the group aesthetic?
```

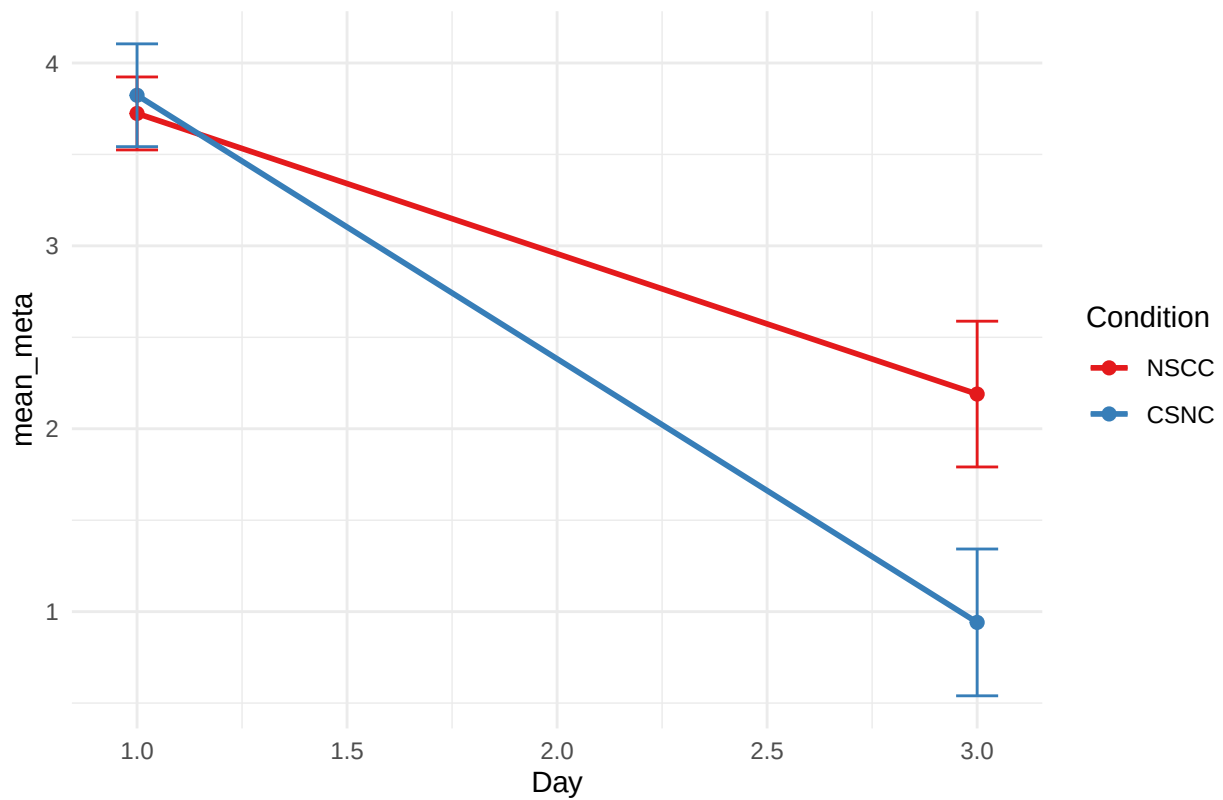


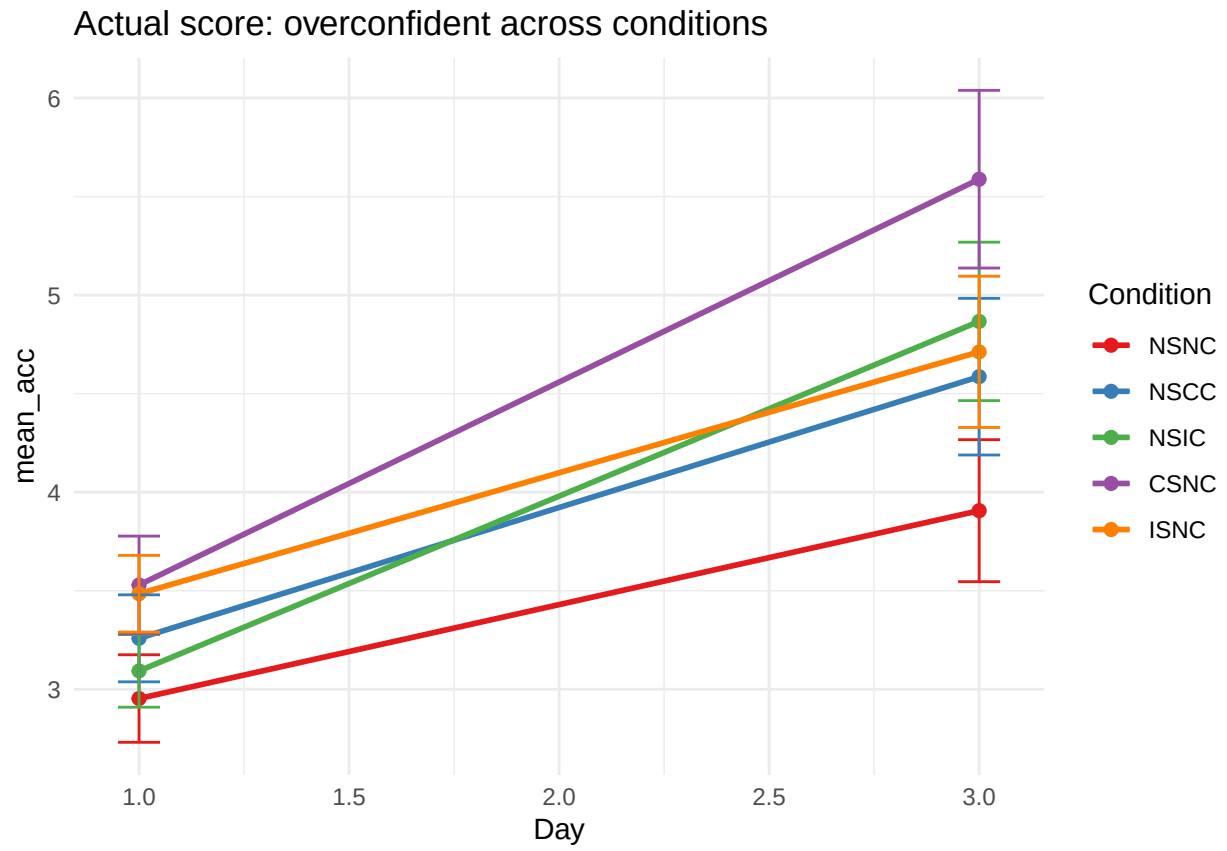
Longitudinal changes

Metacognition: overconfident across conditions

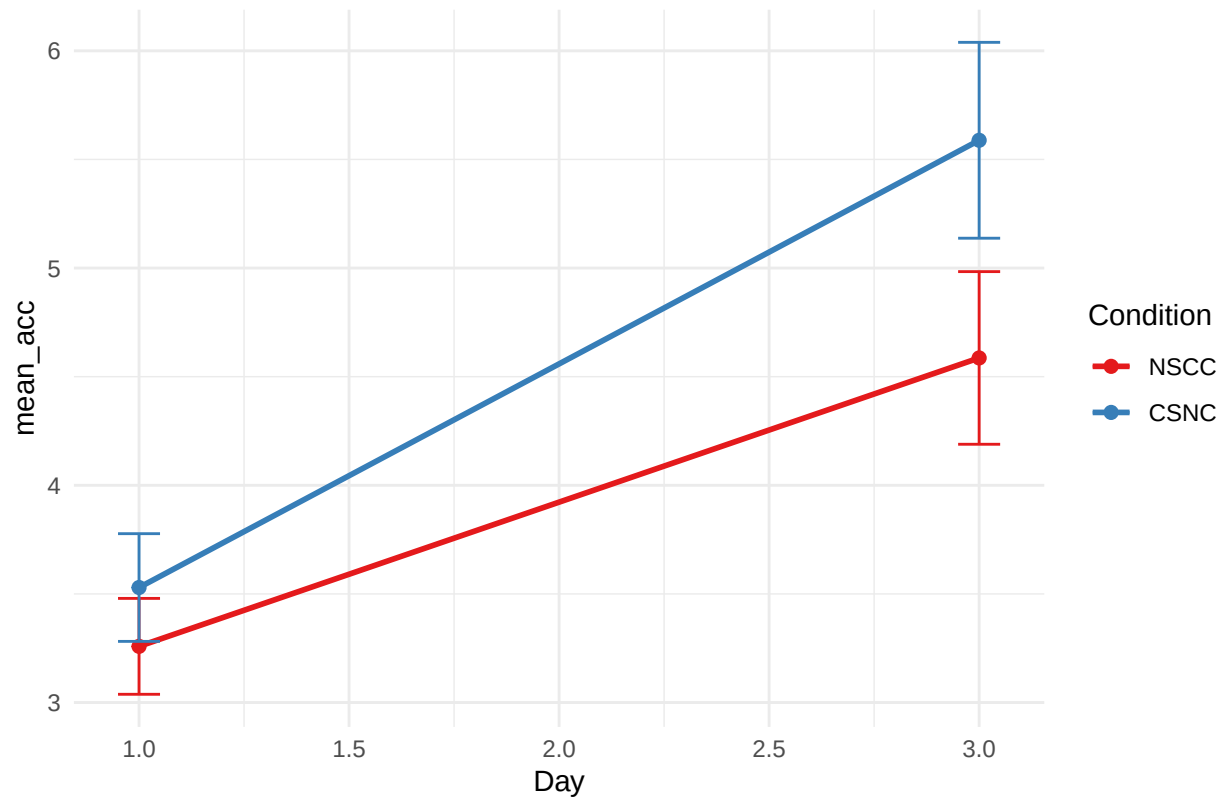


Metacognition: overconfident in NSCC, CSNC

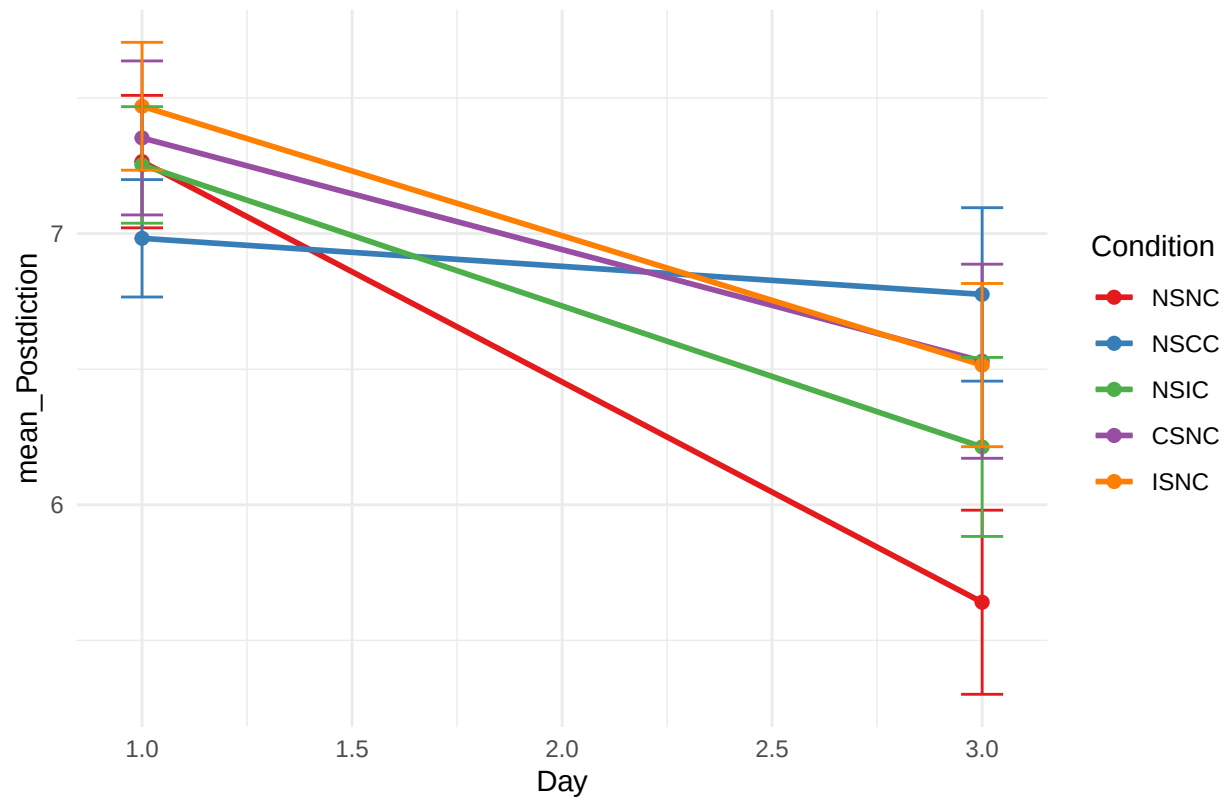




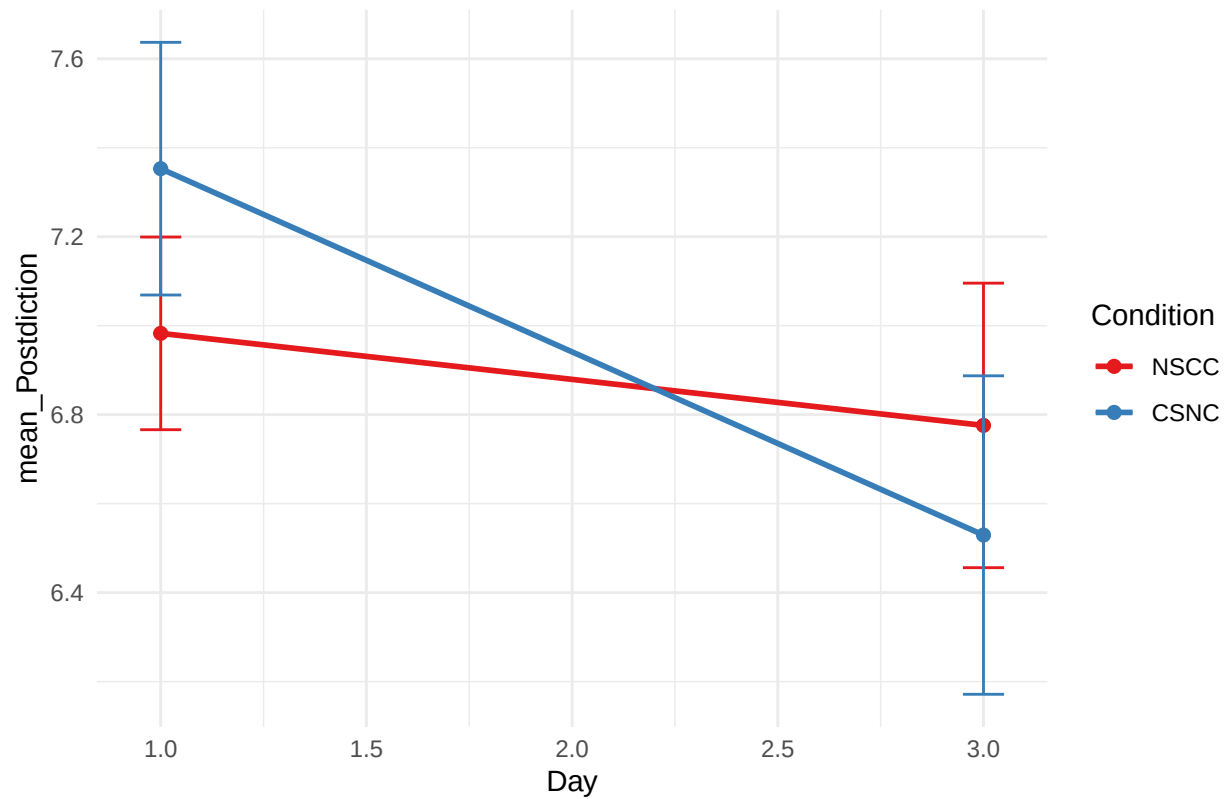
Actual score: overconfident in NSCC, CSNC



Postdiction: overconfident across conditions



Postdiction: overconfident in NSCC, CSNC



Metacognition change visualization

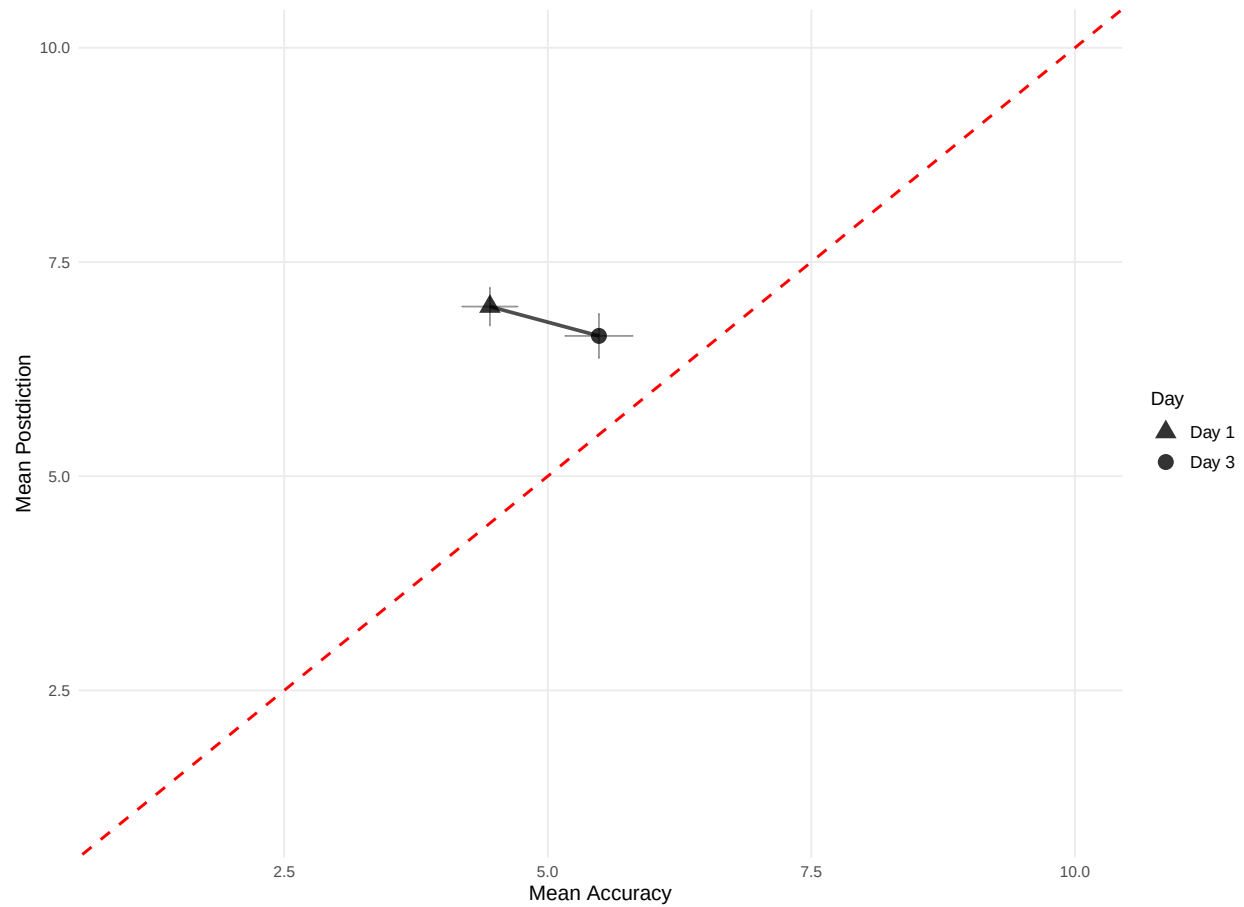
Sample-wide changes

```
## Warning: 'geom_errorbarh()' was deprecated in ggplot2 4.0.0.  
## i Please use the 'orientation' argument of 'geom_errorbar()' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

```
## 'height' was translated to 'width'.
```

Mean Accuracy vs Postdiction by Metacognition on pretest

Triangles = Day 1, Circles = Day 3, Lines connect same conditions
Error bars show 95% confidence intervals



'height' was translated to 'width'.

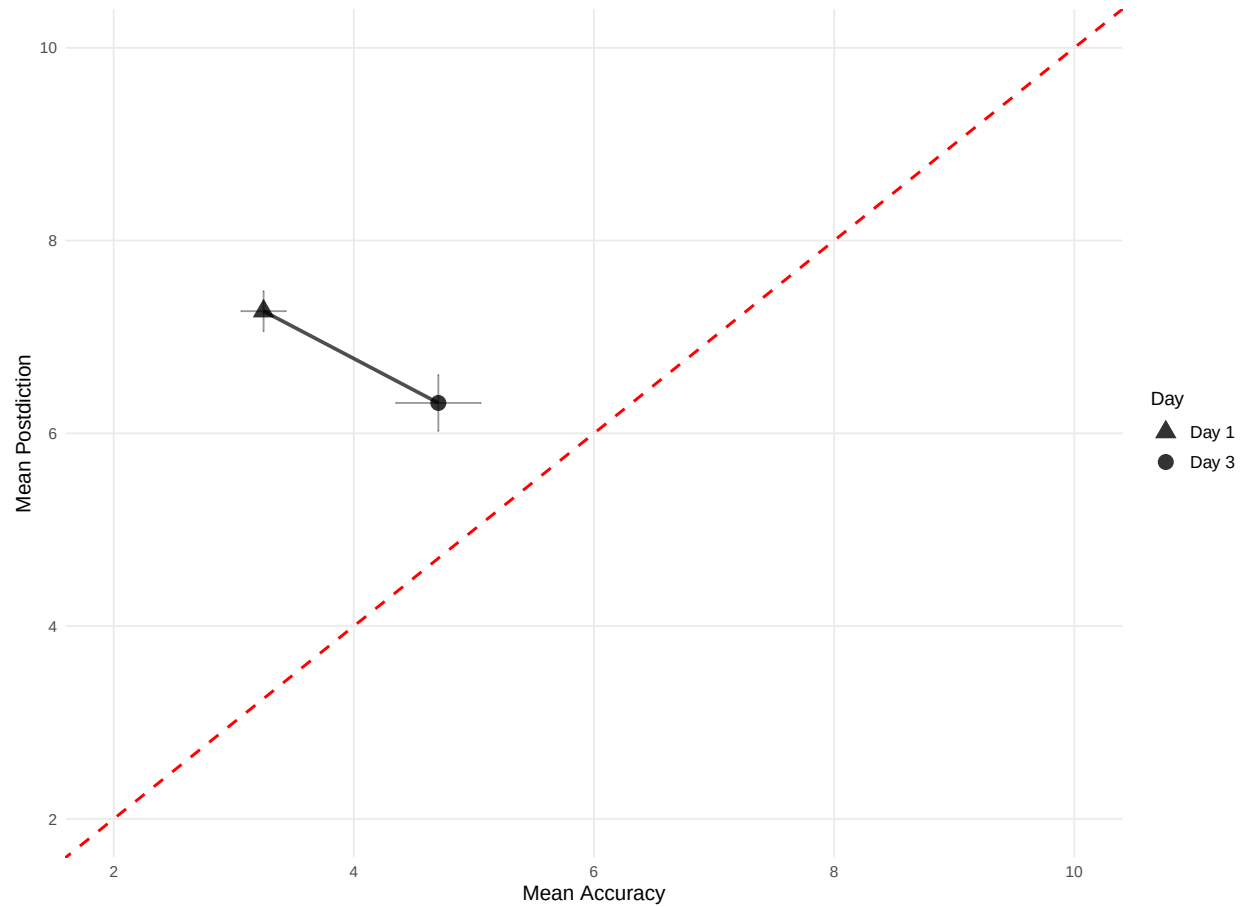
Dif meta groups changes

Over-estimating students' changes

'height' was translated to 'width'.

Overestimating students: Mean Accuracy vs Postdiction by Condition and Day

Triangles = Day 1, Circles = Day 3, Lines connect same conditions
Error bars show 95% confidence intervals



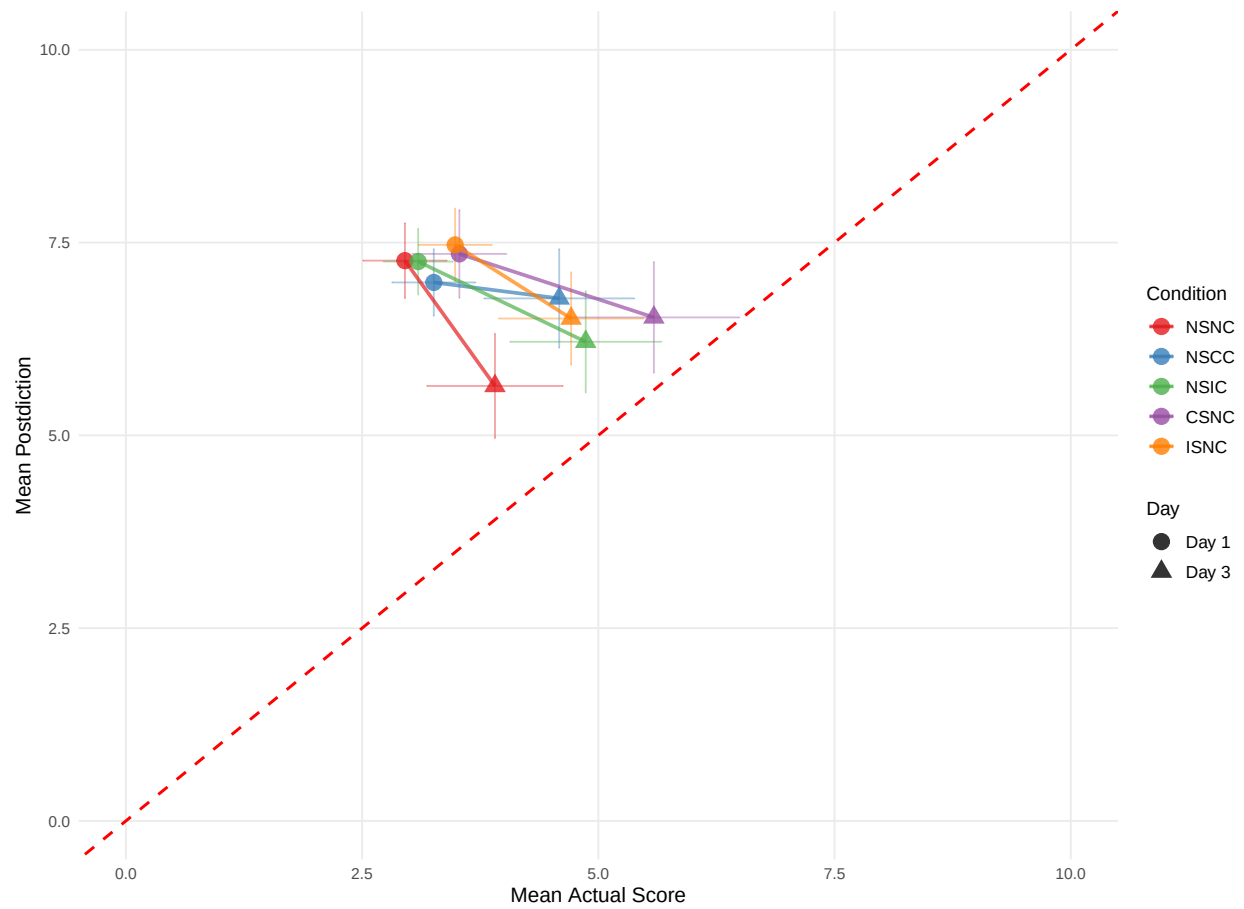
'height' was translated to 'width'.

By condition

'height' was translated to 'width'.

Over-estimating students: Mean Actual Score vs Postdiction by Condition and Day

Triangles = Day 1, Circles = Day 3, Lines connect same conditions
Error bars show 95% confidence intervals



```
## 'height' was translated to 'width'.
```

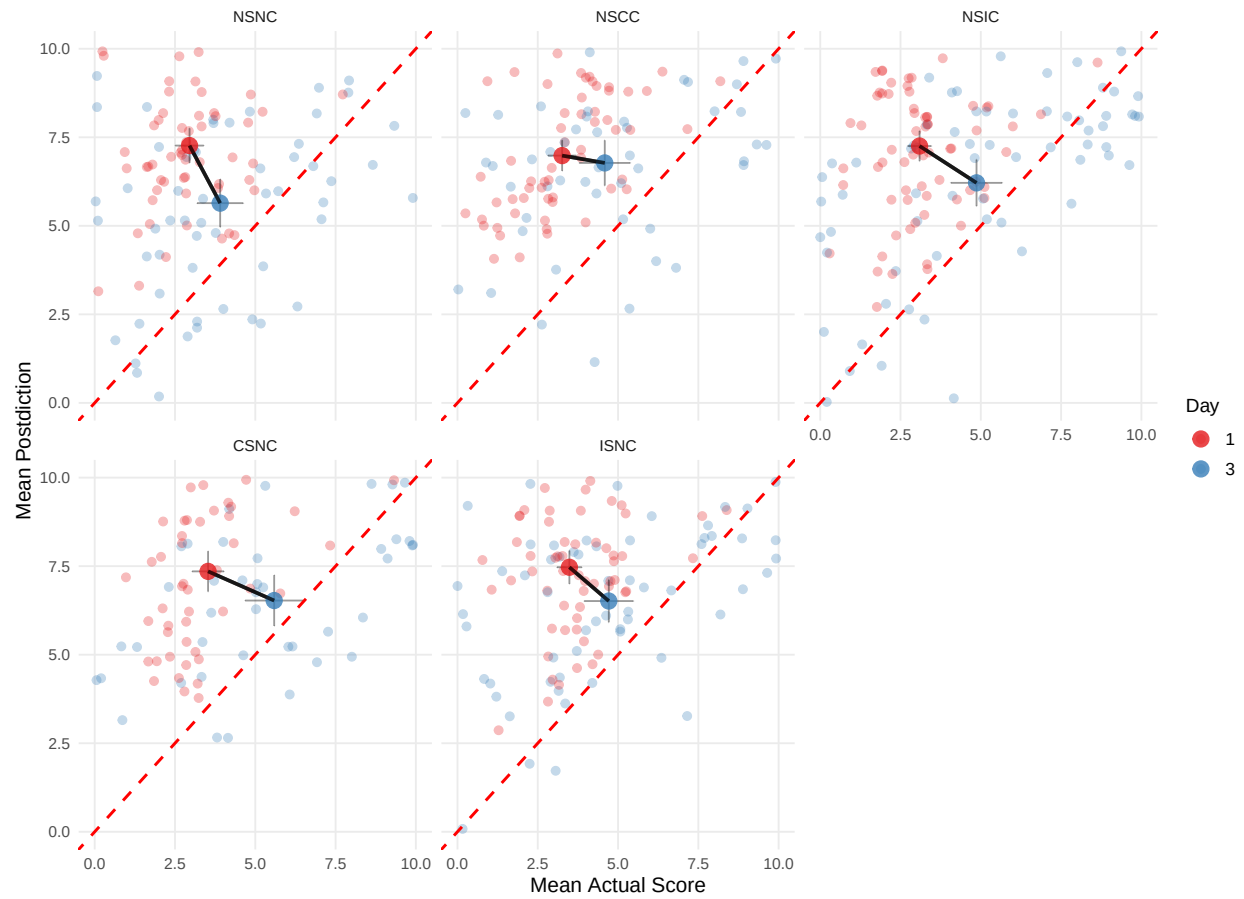
```
## 'height' was translated to 'width'.
```

```
## Warning: Removed 82 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```


Over-estimating students: Mean Actual Score vs Postdiction by Condition and Day

Large shapes = means, small hollow shapes = individual participants
Error bars show 95% confidence intervals

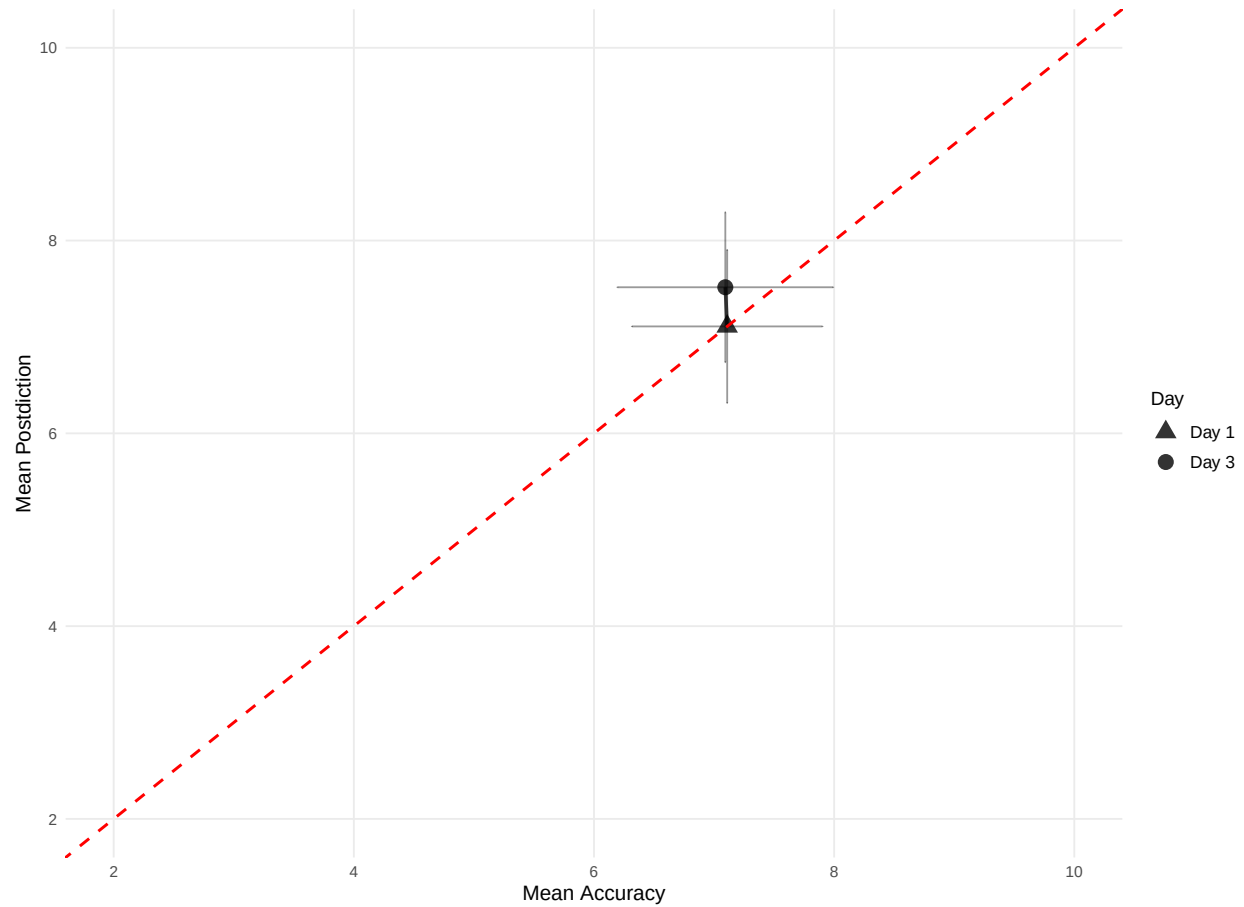


Mid-estimating students' students' changes

'height' was translated to 'width'.

Mid-estimating students: Mean Accuracy vs Postdiction by Condition and Day

Triangles = Day 1, Circles = Day 3, Lines connect same conditions
Error bars show 95% confidence intervals



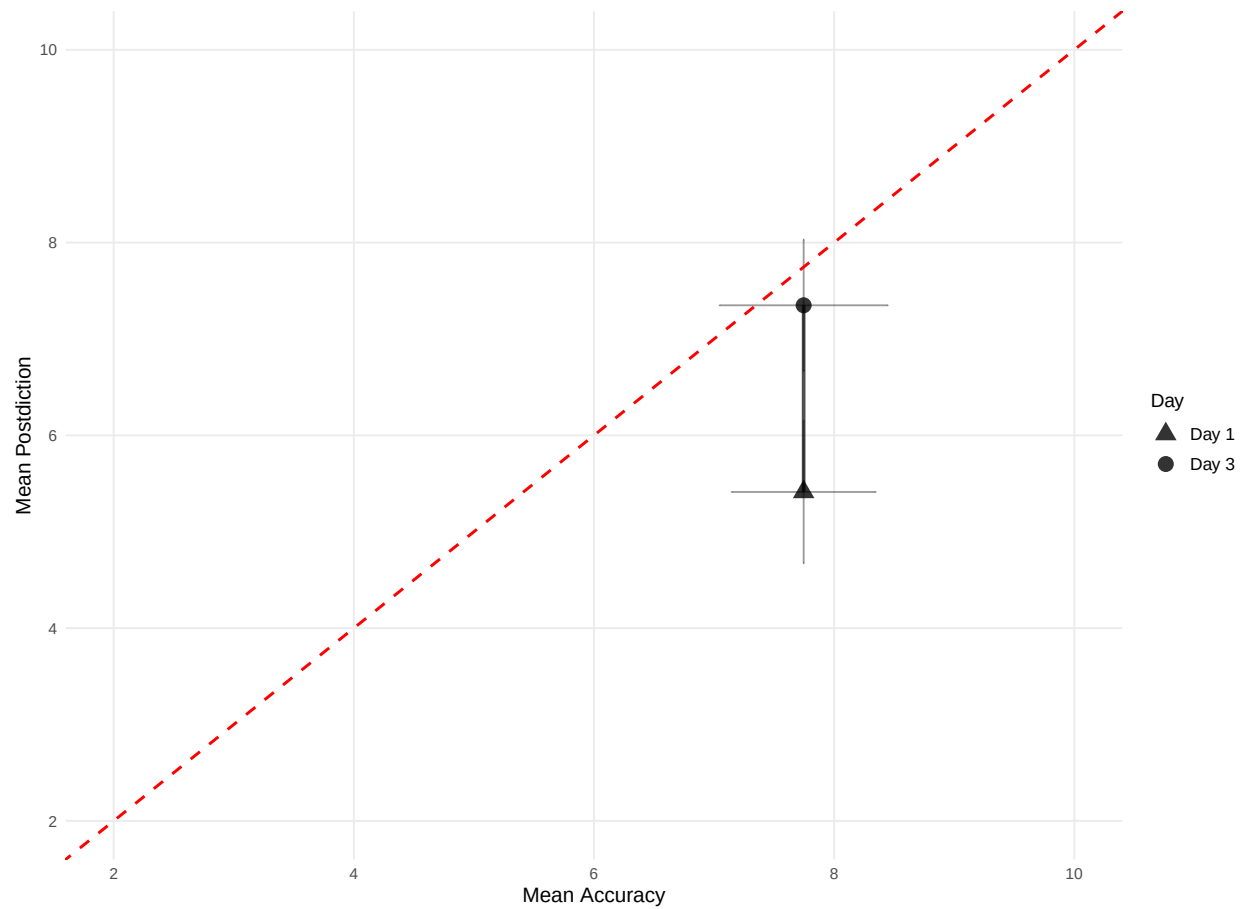
'height' was translated to 'width'.

Under-estimating students'students' changes

'height' was translated to 'width'.

Under-estimating students: Mean Accuracy vs Postdiction by Condition and Day

Triangles = Day 1, Circles = Day 3, Lines connect same conditions
Error bars show 95% confidence intervals



'height' was translated to 'width'.

Statistical comparison

NSNC

```
set.seed(123)

# Metacognition
t.test(over_NSNC$Day1_Metacognition,
       over_NSNC$Day3_Metacognition,
       paired = TRUE, exact = FALSE, conf.int = TRUE)

##
## Paired t-test
##
## data: over_NSNC$Day1_Metacognition and over_NSNC$Day3_Metacognition
## t = 5.8406, df = 63, p-value = 1.977e-07
```

```
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  1.696036 3.460214
## sample estimates:
## mean difference
##      2.578125
```

```
cohens_d(y = over_NSNC$Day3_Metacognition,
         x = over_NSNC$Day1_Metacognition, paired = TRUE)
```

```
## For paired samples, 'repeated_measures_d()' provides more options.
```

```
## Cohen's d |          95% CI
## -----
## 0.73      | [0.45, 1.00]
```

```
# Actual score
t.test(over_NSNC$Day1_Acc,
       over_NSNC$Day3_Acc,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_NSNC$Day1_Acc and over_NSNC$Day3_Acc
## t = -2.863, df = 63, p-value = 0.005691
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -1.61839 -0.28786
## sample estimates:
## mean difference
##      -0.953125
```

```
cohens_d(x = over_NSNC$Day1_Acc,
         y = over_NSNC$Day3_Acc,
         paired = TRUE)
```

```
## For paired samples, 'repeated_measures_d()' provides more options.
```

```
## Cohen's d |          95% CI
## -----
## -0.36     | [-0.61, -0.10]
```

```
# Postdiction
t.test(over_NSNC$Day1_Postdiction,
       over_NSNC$Day3_Postdiction,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
```

```
##
## data: over_NSNC$Day1_Postdiction and over_NSNC$Day3_Postdiction
## t = 4.4281, df = 63, p-value = 3.86e-05
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 0.8916536 2.3583464
## sample estimates:
## mean difference
## 1.625
```

```
cohens_d(x = over_NSNC$Day1_Postdiction,
         y = over_NSNC$Day3_Postdiction,
         paired = TRUE)
```

```
## For paired samples, 'repeated_measures_d()' provides more options.
```

```
## Cohen's d |          95% CI
## -----
## 0.55      | [0.29, 0.81]
```

```
# Correlation between actual score and postdiction before intervention
cor.test(over_NSNC$Day1_Acc,
         over_NSNC$Day1_Postdiction)
```

```
##
## Pearson's product-moment correlation
##
## data: over_NSNC$Day1_Acc and over_NSNC$Day1_Postdiction
## t = 2.3196, df = 62, p-value = 0.02367
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.03952159 0.49407535
## sample estimates:
## cor
## 0.2825856
```

```
# Correlation between actual score and postdiction after intervention
cor.test(over_NSNC$Day3_Acc,
         over_NSNC$Day3_Postdiction)
```

```
##
## Pearson's product-moment correlation
##
## data: over_NSNC$Day3_Acc and over_NSNC$Day3_Postdiction
## t = 3.595, df = 62, p-value = 0.0006433
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.1887892 0.5998908
## sample estimates:
## cor
## 0.4153248
```

ISNC

```
# Metacognition
t.test(over_ISNC$Day1_Metacognition,
       over_ISNC$Day3_Metacognition,
       paired=T, exact = FALSE, conf.int = TRUE)

##
## Paired t-test
##
## data: over_ISNC$Day1_Metacognition and over_ISNC$Day3_Metacognition
## t = 5.053, df = 65, p-value = 3.775e-06
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  1.319485 3.044151
## sample estimates:
## mean difference
##      2.181818
```

```
cohens_d(x = over_ISNC$Day1_Metacognition,
         y = over_ISNC$Day3_Metacognition,
         paired = TRUE)
```

For paired samples, 'repeated_measures_d()' provides more options.

```
## Cohen's d |      95% CI
## -----
## 0.62      | [0.36, 0.88]
```

```
# Actual score
t.test(over_ISNC$Day1_Acc,
       over_ISNC$Day3_Acc,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_ISNC$Day1_Acc and over_ISNC$Day3_Acc
## t = -3.4734, df = 65, p-value = 0.0009195
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -1.9329336 -0.5216119
## sample estimates:
## mean difference
##      -1.227273
```

```
cohens_d(x = over_ISNC$Day1_Acc,
         y = over_ISNC$Day3_Acc,
         paired = TRUE)
```

For paired samples, 'repeated_measures_d()' provides more options.

```
## Cohen's d |          95% CI
## -----
## -0.43      | [-0.68, -0.17]
```

```
# Postdiction
t.test(over_ISNC$Day1_Postdiction,
       over_ISNC$Day3_Postdiction,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_ISNC$Day1_Postdiction and over_ISNC$Day3_Postdiction
## t = 2.5011, df = 65, p-value = 0.01491
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  0.192340 1.716751
## sample estimates:
## mean difference
##      0.9545455
```

```
cohens_d(x = over_ISNC$Day1_Postdiction,
         y = over_ISNC$Day3_Postdiction,
         paired = TRUE)
```

For paired samples, 'repeated_measures_d()' provides more options.

```
## Cohen's d |          95% CI
## -----
## 0.31       | [0.06, 0.55]
```

```
# Correlation between actual score and postdiction before intervention
cor.test(over_ISNC$Day1_Acc,
         over_ISNC$Day1_Postdiction)
```

```
##
## Pearson's product-moment correlation
##
## data: over_ISNC$Day1_Acc and over_ISNC$Day1_Postdiction
## t = 1.4443, df = 64, p-value = 0.1535
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.0672623 0.4023926
## sample estimates:
##      cor
## 0.1776628
```

```
# Correlation between actual score and postdiction after intervention
cor.test(over_ISNC$Day3_Acc,
         over_ISNC$Day3_Postdiction)
```

```
##
## Pearson's product-moment correlation
##
## data: over_ISNC$Day3_Acc and over_ISNC$Day3_Postdiction
## t = 4.0288, df = 64, p-value = 0.0001513
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2331346 0.6239004
## sample estimates:
##      cor
## 0.4497875
```

CSNC

```
# Metacognition
t.test(over_CSNC$Day1_Metacognition,
       over_CSNC$Day3_Metacognition,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_CSNC$Day1_Metacognition and over_CSNC$Day3_Metacognition
## t = 6.7046, df = 50, p-value = 1.74e-08
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  2.018857 3.745849
## sample estimates:
## mean difference
##      2.882353
```

```
cohens_d(x = over_CSNC$Day1_Metacognition,
         y = over_CSNC$Day3_Metacognition,
         paired = TRUE)
```

For paired samples, 'repeated_measures_d()' provides more options.

```
## Cohen's d |          95% CI
## -----
## 0.94      | [0.61, 1.27]
```

```
# Actual score
t.test(over_CSNC$Day1_Acc,
       over_CSNC$Day3_Acc,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_CSNC$Day1_Acc and over_CSNC$Day3_Acc
```



```
## t = -5.1169, df = 50, p-value = 4.958e-06
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -2.866982 -1.250665
## sample estimates:
## mean difference
## -2.058824
```

```
cohens_d(x = over_CSNC$Day1_Acc,
          y = over_CSNC$Day3_Acc,
          paired = TRUE)
```

For paired samples, 'repeated_measures_d()' provides more options.

```
## Cohen's d |          95% CI
## -----
## -0.72      | [-1.02, -0.41]
```

```
# Postdiction
t.test(over_CSNC$Day1_Postdiction,
        over_CSNC$Day3_Postdiction,
        paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_CSNC$Day1_Postdiction and over_CSNC$Day3_Postdiction
## t = 2.4702, df = 50, p-value = 0.01696
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 0.153917 1.493142
## sample estimates:
## mean difference
## 0.8235294
```

```
cohens_d(x = over_CSNC$Day1_Postdiction,
          y = over_CSNC$Day3_Postdiction,
          paired = TRUE)
```

For paired samples, 'repeated_measures_d()' provides more options.

```
## Cohen's d |          95% CI
## -----
## 0.35      | [0.06, 0.63]
```

```
# Correlation between actual score and postdiction before intervention
cor.test(over_CSNC$Day1_Acc,
          over_CSNC$Day1_Postdiction)
```

```
##
```

```
## Pearson's product-moment correlation
##
## data: over_CSNC$Day1_Acc and over_CSNC$Day1_Postdiction
## t = 3.5103, df = 49, p-value = 0.0009709
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.1970182 0.6442591
## sample estimates:
## cor
## 0.4482634
```

```
# Correlation between actual score and postdiction after intervention
cor.test(over_CSNC$Day3_Acc,
          over_CSNC$Day3_Postdiction)
```

```
##
## Pearson's product-moment correlation
##
## data: over_CSNC$Day3_Acc and over_CSNC$Day3_Postdiction
## t = 4.3475, df = 49, p-value = 6.939e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.2948918 0.7012260
## sample estimates:
## cor
## 0.5275989
```

NSCC

```
# Metacognition
t.test(over_NSCC$Day1_Metacognition,
        over_NSCC$Day3_Metacognition,
        paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_NSCC$Day1_Metacognition and over_NSCC$Day3_Metacognition
## t = 4.2726, df = 57, p-value = 7.418e-05
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 0.8153021 2.2536634
## sample estimates:
## mean difference
## 1.534483
```

```
cohens_d(x = over_NSCC$Day1_Metacognition,
          y = over_NSCC$Day3_Metacognition,
          paired = TRUE)
```

```
## For paired samples, 'repeated_measures_d()' provides more options.
```

```
## Cohen's d |          95% CI
## -----
## 0.56      | [0.28, 0.84]
```

```
# Actual score
t.test(over_NSCC$Day1_Acc,
       over_NSCC$Day3_Acc,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_NSCC$Day1_Acc and over_NSCC$Day3_Acc
## t = -4.111, df = 57, p-value = 0.0001278
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -1.9742552 -0.6809172
## sample estimates:
## mean difference
##      -1.327586
```

```
cohens_d(x = over_NSCC$Day1_Acc,
         y = over_NSCC$Day3_Acc,
         paired = TRUE)
```

For paired samples, 'repeated_measures_d()' provides more options.

```
## Cohen's d |          95% CI
## -----
## -0.54      | [-0.81, -0.26]
```

```
# Postdiction
t.test(over_NSCC$Day1_Postdiction,
       over_NSCC$Day3_Postdiction,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_NSCC$Day1_Postdiction and over_NSCC$Day3_Postdiction
## t = 0.80292, df = 57, p-value = 0.4254
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -0.3091013  0.7228944
## sample estimates:
## mean difference
##      0.2068966
```

```
cohens_d(x = over_NSCC$Day1_Postdiction,
         y = over_NSCC$Day3_Postdiction,
         paired = TRUE)
```

```
## For paired samples, 'repeated_measures_d()' provides more options.
```

```
## Cohen's d |          95% CI  
## -----  
## 0.11      | [-0.15, 0.36]
```

```
# Correlation between actual score and postdiction before intervention  
cor.test(over_NSCC$Day1_Acc,  
          over_NSCC$Day1_Postdiction)
```

```
##  
## Pearson's product-moment correlation  
##  
## data: over_NSCC$Day1_Acc and over_NSCC$Day1_Postdiction  
## t = 5.3814, df = 56, p-value = 1.505e-06  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## 0.3833482 0.7317767  
## sample estimates:  
## cor  
## 0.5838343
```

```
# Correlation between actual score and postdiction after intervention  
cor.test(over_NSCC$Day3_Acc,  
          over_NSCC$Day3_Postdiction)
```

```
##  
## Pearson's product-moment correlation  
##  
## data: over_NSCC$Day3_Acc and over_NSCC$Day3_Postdiction  
## t = 3.2561, df = 56, p-value = 0.00192  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## 0.1568469 0.5958678  
## sample estimates:  
## cor  
## 0.3989789
```

NSIC

```
# Metacognition  
t.test(over_NSIC$Day1_Metacognition,  
        over_NSIC$Day3_Metacognition,  
        paired=T, exact = FALSE, conf.int = TRUE)
```

```
##  
## Paired t-test  
##  
## data: over_NSIC$Day1_Metacognition and over_NSIC$Day3_Metacognition  
## t = 9.1788, df = 74, p-value = 7.649e-14
```

```
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  2.202615 3.424052
## sample estimates:
## mean difference
##      2.813333
```

```
cohens_d(x = over_NSIC$Day1_Metacognition,
         y = over_NSIC$Day3_Metacognition,
         paired = TRUE)
```

```
## For paired samples, 'repeated_measures_d()' provides more options.
```

```
## Cohen's d |          95% CI
## -----
## 1.06      | [0.77, 1.34]
```

```
# Actual score
t.test(over_NSIC$Day1_Acc,
       over_NSIC$Day3_Acc,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
## Paired t-test
##
## data: over_NSIC$Day1_Acc and over_NSIC$Day3_Acc
## t = -5.3527, df = 74, p-value = 9.338e-07
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -2.433452 -1.113214
## sample estimates:
## mean difference
##      -1.773333
```

```
cohens_d(x = over_NSIC$Day1_Acc,
         y = over_NSIC$Day3_Acc,
         paired = TRUE)
```

```
## For paired samples, 'repeated_measures_d()' provides more options.
```

```
## Cohen's d |          95% CI
## -----
## -0.62     | [-0.86, -0.37]
```

```
# Postdiction
t.test(over_NSIC$Day1_Postdiction,
       over_NSIC$Day3_Postdiction,
       paired=T, exact = FALSE, conf.int = TRUE)
```

```
##
```

```
## Paired t-test
##
## data: over_NSIC$Day1_Postdiction and over_NSIC$Day3_Postdiction
## t = 3.4211, df = 74, p-value = 0.001018
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 0.4342838 1.6457162
## sample estimates:
## mean difference
## 1.04
```

```
cohens_d(x = over_NSIC$Day1_Postdiction,
         y = over_NSIC$Day3_Postdiction,
         paired = TRUE)
```

```
## For paired samples, 'repeated_measures_d()' provides more options.
```

```
## Cohen's d | 95% CI
## -----
## 0.40 | [0.16, 0.63]
```

```
# Correlation between actual score and postdiction before intervention
cor.test(over_NSIC$Day1_Acc,
         over_NSIC$Day1_Postdiction)
```

```
##
## Pearson's product-moment correlation
##
## data: over_NSIC$Day1_Acc and over_NSIC$Day1_Postdiction
## t = 2.1339, df = 73, p-value = 0.03621
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.01624177 0.44480982
## sample estimates:
## cor
## 0.2423104
```

```
# Correlation between actual score and postdiction after intervention
cor.test(over_NSIC$Day3_Acc,
         over_NSIC$Day3_Postdiction)
```

```
##
## Pearson's product-moment correlation
##
## data: over_NSIC$Day3_Acc and over_NSIC$Day3_Postdiction
## t = 6.9878, df = 73, p-value = 1.097e-09
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.4742753 0.7519990
## sample estimates:
## cor
## 0.6330899
```

Anova

Overestimation change

```
# Run one-way ANOVA
anova_result <- aov(Meta_Change ~ Condition, data = only_overestimating)
summary(anova_result)

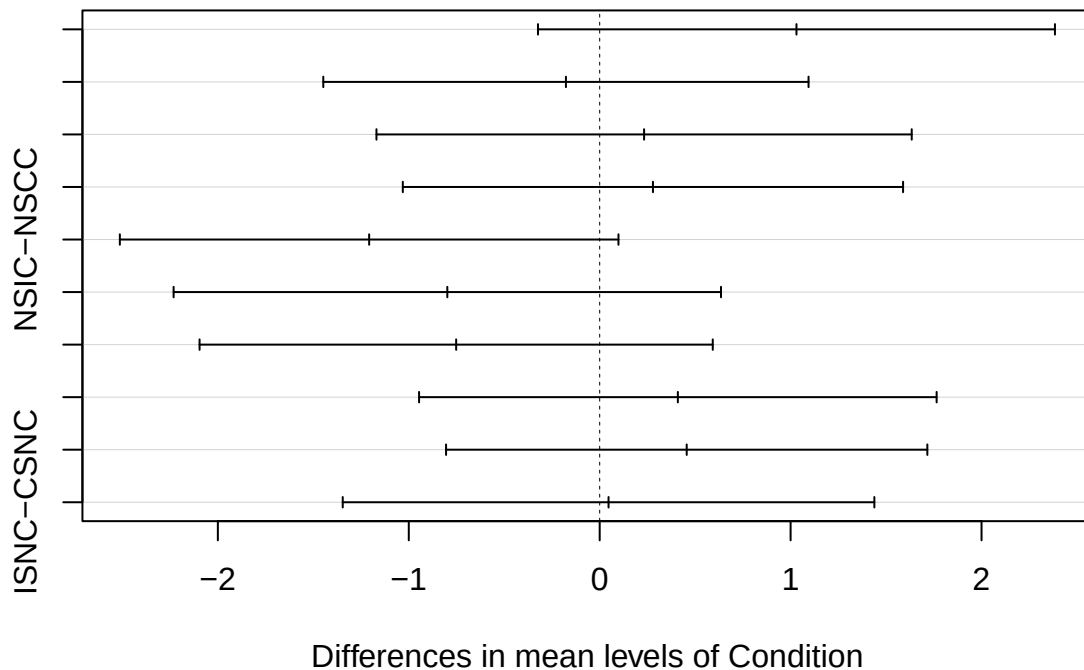
##              Df Sum Sq Mean Sq F value Pr(>F)
## Condition      4   53.1   13.270   1.791   0.13
## Residuals    309 2288.9    7.407

# Tukey HSD post-hoc test for pairwise comparisons
pairwise <- TukeyHSD(anova_result)
print(pairwise)

##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = Meta_Change ~ Condition, data = only_overestimating)
##
## $Condition
##              diff              lwr              upr              p adj
## NSCC-NSNC  1.03071121 -0.3231846  2.38460706  0.2274513
## NSIC-NSNC -0.17687500 -1.4477303  1.09398031  0.9954710
## CSNC-NSNC  0.23253676 -1.1692539  1.63432745  0.9911175
## ISNC-NSNC  0.27888258 -1.0312615  1.58902662  0.9773643
## NSIC-NSCC -1.20758621 -2.5134279  0.09825545  0.0851111
## CSNC-NSCC -0.79817444 -2.2317596  0.63541070  0.5452088
## ISNC-NSCC -0.75182863 -2.0959367  0.59227949  0.5406038
## CSNC-NSIC  0.40941176 -0.9460238  1.76484734  0.9215580
## ISNC-NSIC  0.45575758 -0.8046653  1.71618047  0.8587600
## ISNC-CSNC  0.04634581 -1.3459939  1.43868549  0.9999842

# Optional: visualize confidence intervals
plot(pairwise)
```

95% family-wise confidence level



Actual score change

```
# Run one-way ANOVA
anova_result <- aov(Acc_Change ~ Condition, data = only_overestimating)
summary(anova_result)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## Condition    4   46.7   11.670    1.535  0.192
## Residuals  309 2349.2    7.603
```

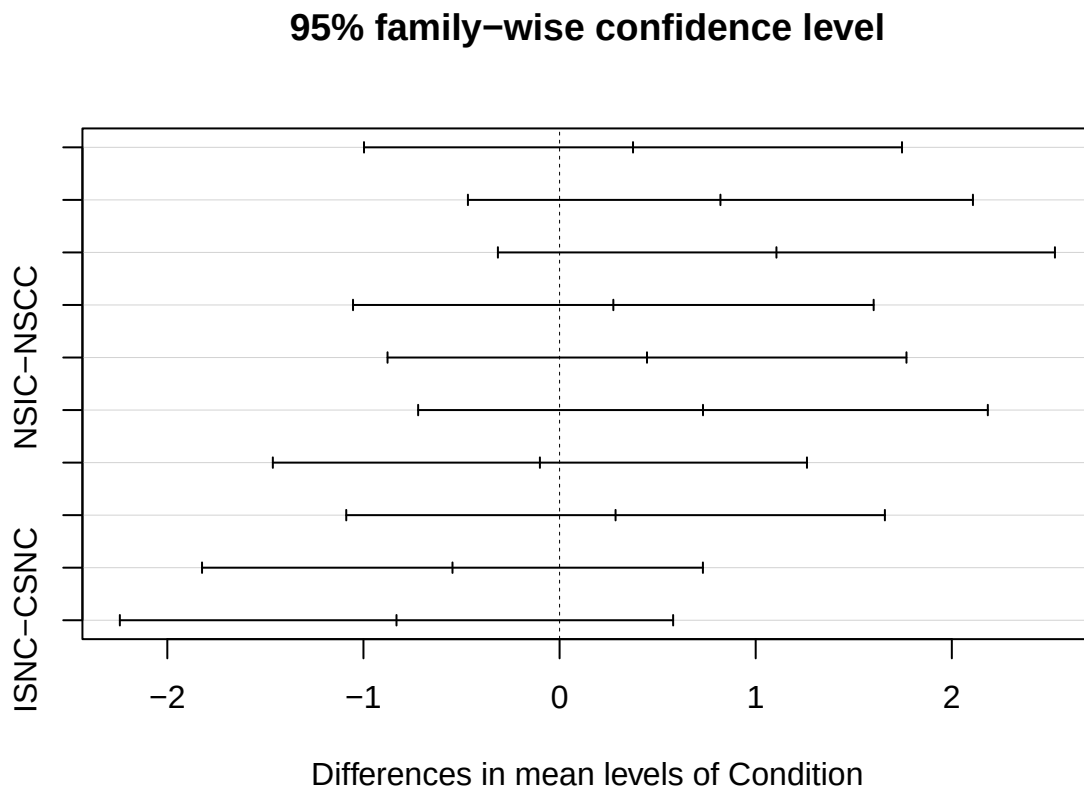
```
# Tukey HSD post-hoc test for pairwise comparisons
pairwise <- TukeyHSD(anova_result)
print(pairwise)
```

```
##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = Acc_Change ~ Condition, data = only_overestimating)
##
## $Condition
##           diff          lwr          upr          p adj
## NSCC-NSNC  0.3744612 -0.9971568 1.7460792 0.9446027
## NSIC-NSNC  0.8202083 -0.4672822 2.1076988 0.4061383
```



```
## CSNC-NSNC 1.1056985 -0.3144413 2.5258383 0.2073791
## ISNC-NSNC 0.2741477 -1.0531458 1.6014412 0.9797491
## NSIC-NSCC 0.4457471 -0.8771877 1.7686819 0.8872755
## CSNC-NSCC 0.7312373 -0.7211131 2.1835877 0.6399840
## ISNC-NSCC -0.1003135 -1.4620156 1.2613887 0.9996266
## CSNC-NSIC 0.2854902 -1.0876877 1.6586681 0.9792517
## ISNC-NSIC -0.5460606 -1.8229821 0.7308609 0.7665812
## ISNC-CSNC -0.8315508 -2.2421159 0.5790143 0.4872386
```

```
# Optional: visualize confidence intervals
plot(pairwise)
```



Postdiction change

```
# Run one-way ANOVA
anova_result <- aov(Postdiction_Change ~ Condition, data = only_overestimating)
summary(anova_result)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## Condition    4   62.6   15.653    2.215 0.0673 .
## Residuals  309 2183.7    7.067
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Tukey HSD post-hoc test for pairwise comparisons
```

```
pairwise <- TukeyHSD(anova_result)
```

```
print(pairwise)
```

```
## Tukey multiple comparisons of means
```

```
## 95% family-wise confidence level
```

```
##
```

```
## Fit: aov(formula = Postdiction_Change ~ Condition, data = only_overestimating)
```

```
##
```

```
## $Condition
```

```
##          diff          lwr          upr      p adj
```

```
## NSCC-NSNC  1.41810345  0.0956899  2.7405170  0.0286644
```

```
## NSIC-NSNC  0.58500000 -0.6563040  1.8263040  0.6956439
```

```
## CSNC-NSNC  0.80147059 -0.5677241  2.1706653  0.4945318
```

```
## ISNC-NSNC  0.67045455 -0.6092246  1.9501337  0.6038154
```

```
## NSIC-NSCC -0.83310345 -2.1085802  0.4423733  0.3799182
```

```
## CSNC-NSCC -0.61663286 -2.0168827  0.7836170  0.7465539
```

```
## ISNC-NSCC -0.74764890 -2.0605023  0.5652045  0.5225803
```

```
## CSNC-NSIC  0.21647059 -1.1074469  1.5403881  0.9915929
```

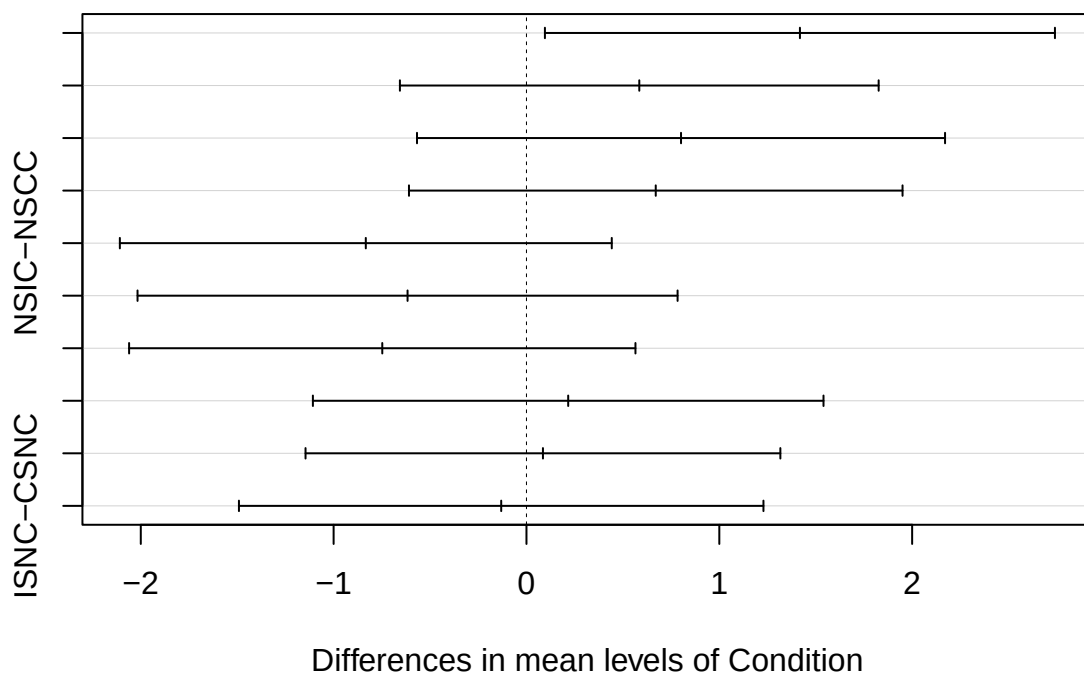
```
## ISNC-NSIC  0.08545455 -1.1456596  1.3165687  0.9997050
```

```
## ISNC-CSNC -0.13101604 -1.4909795  1.2289474  0.9989226
```

```
# Optional: visualize confidence intervals
```

```
plot(pairwise)
```

95% family-wise confidence level



Linear models

```
## boundary (singular) fit: see help('isSingular')

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Meta_Change ~ Condition + (1 | TeacherUniqueID)
## Data: only_overestimating
##
## REML criterion at convergence: 1516.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.68105 -0.57897 -0.04409  0.62577  3.09527
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## TeacherUniqueID (Intercept) 0.000    0.000
## Residual                7.407    2.722
## Number of obs: 314, groups: TeacherUniqueID, 16
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -1.7031    0.3402 309.0000  -5.006 9.35e-07 ***
## ConditionNSCC    1.0307    0.4934 309.0000   2.089  0.0375 *
## ConditionNSIC   -0.1769    0.4631 309.0000  -0.382  0.7028
## ConditionCSNC    0.2325    0.5109 309.0000   0.455  0.6493
## ConditionISNC    0.2789    0.4775 309.0000   0.584  0.5596
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) CnNSCC CnNSIC CnCSNC
## ConditnNSCC -0.689
## ConditnNSIC -0.735  0.506
## ConditnCSNC -0.666  0.459  0.489
## ConditnISNC -0.713  0.491  0.523  0.475
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Acc_Change ~ Condition + (1 | TeacherUniqueID)
## Data: only_overestimating
##
## REML criterion at convergence: 1521.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.8874 -0.6576 -0.1431  0.6337  2.3461
##
## Random effects:
## Groups           Name          Variance Std.Dev.
```

```

## TeacherUniqueID (Intercept) 0.335    0.5788
## Residual                    7.315    2.7046
## Number of obs: 314, groups: TeacherUniqueID, 16
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    0.8747    0.3739  81.7964   2.340   0.0217 *
## ConditionNSCC    0.2925    0.4944 304.2945   0.592   0.5545
## ConditionNSIC    0.8530    0.4634 302.8669   1.841   0.0666 .
## ConditionCSNC    1.2088    0.5133 306.3526   2.355   0.0192 *
## ConditionISNC    0.2634    0.4783 304.3510   0.551   0.5823
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) CnNSCC CnNSIC CnCSNC
## ConditnNSCC -0.617
## ConditnNSIC -0.661  0.498
## ConditnCSNC -0.608  0.452  0.493
## ConditnISNC -0.644  0.494  0.522  0.473

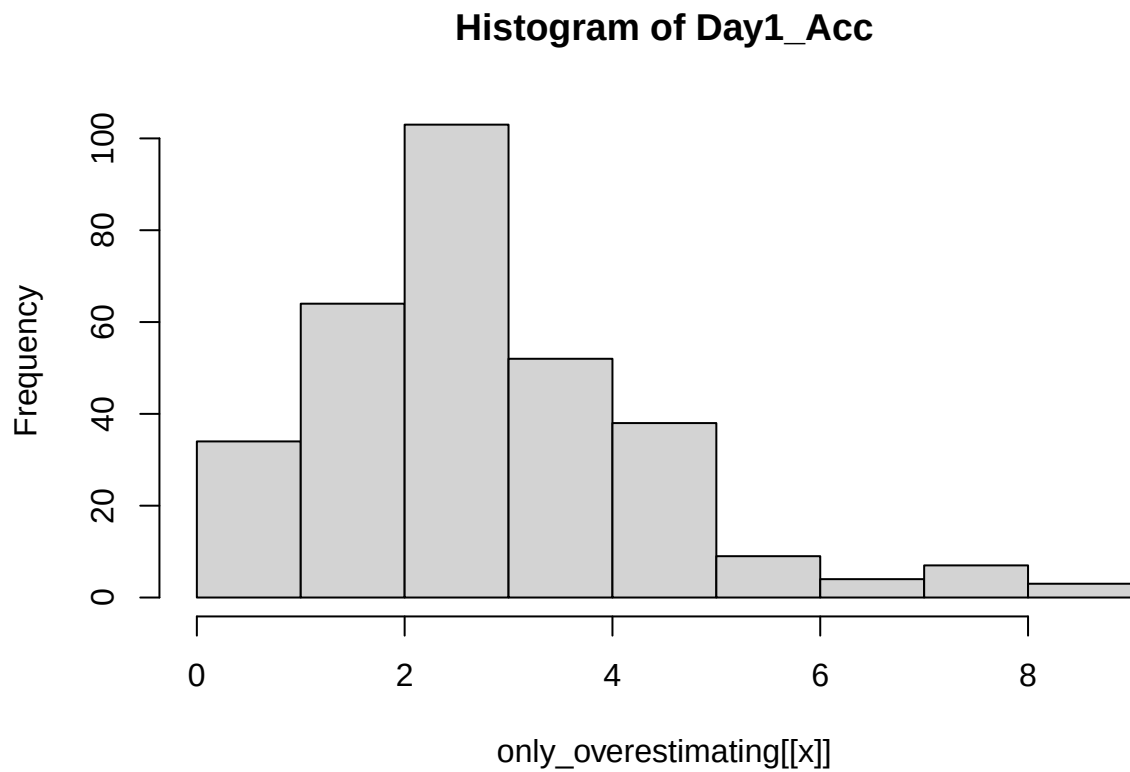
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Postdiction_Change ~ Condition + (1 | TeacherUniqueID)
## Data: only_overestimating
##
## REML criterion at convergence: 1498.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.4720 -0.5677  0.1286  0.5971  3.6987
##
## Random effects:
## Groups           Name      Variance Std.Dev.
## TeacherUniqueID (Intercept) 0.3169   0.5629
## Residual                6.7993   2.6075
## Number of obs: 314, groups: TeacherUniqueID, 16
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -1.6567    0.3610  80.3408  -4.590 1.62e-05 ***
## ConditionNSCC    1.3988    0.4767 304.1856   2.934   0.0036 **
## ConditionNSIC    0.6001    0.4468 302.7371   1.343   0.1802
## ConditionCSNC    0.8455    0.4949 306.2612   1.708   0.0886 .
## ConditionISNC    0.6908    0.4612 304.2458   1.498   0.1352
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) CnNSCC CnNSIC CnCSNC
## ConditnNSCC -0.616
## ConditnNSIC -0.660  0.498
## ConditnCSNC -0.607  0.452  0.493
## ConditnISNC -0.643  0.494  0.522  0.473

```

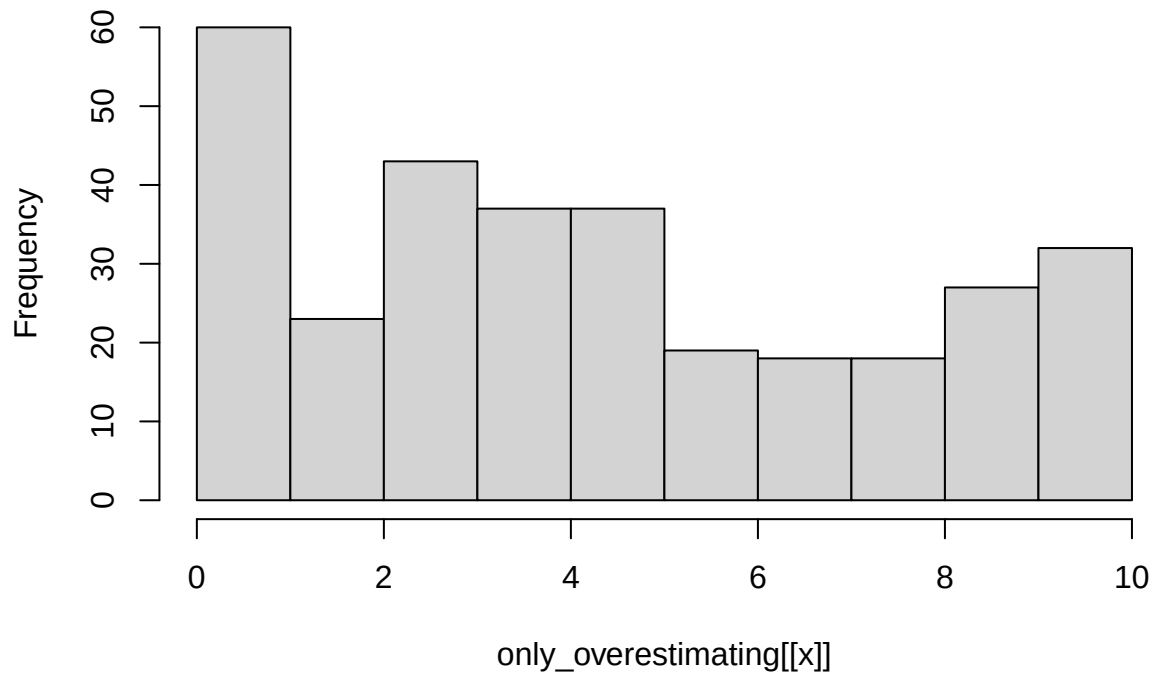
Descr plots for overestimating kids

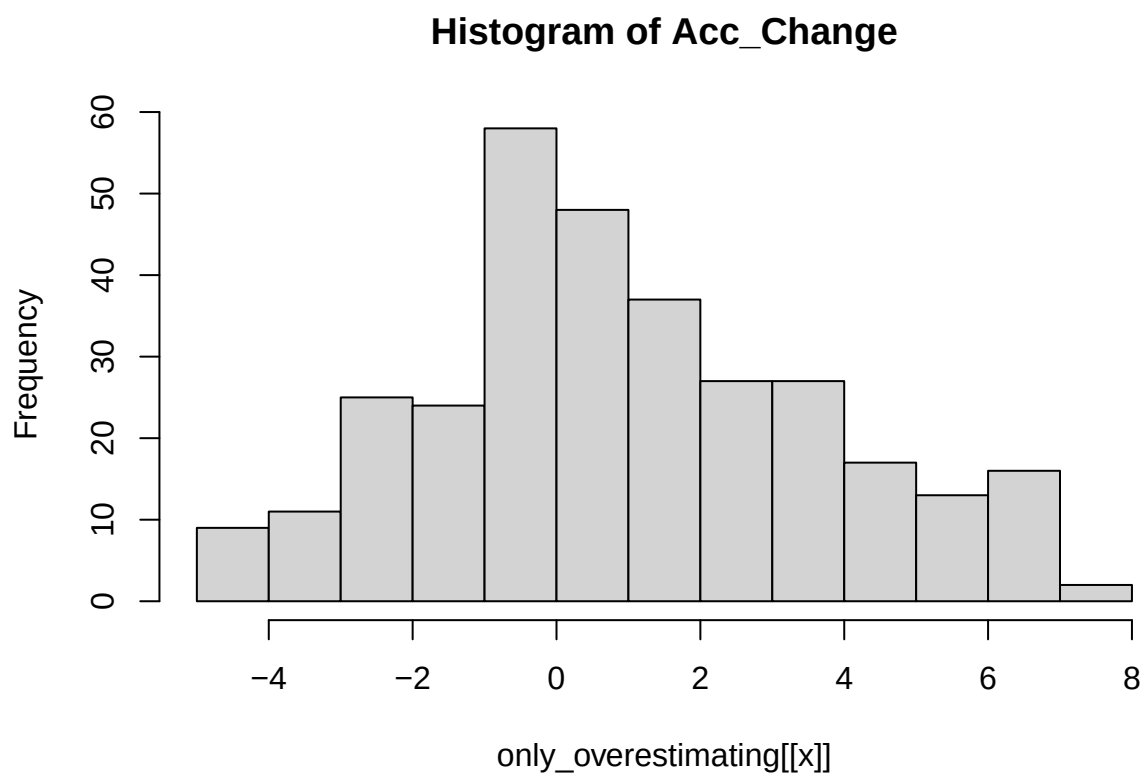
All accuracies

```
lapply(  
  c("Day1_Acc", "Day3_Acc", "Acc_Change"),  
  function(x)  
    hist(only_overestimating[[x]], main = paste("Histogram of", x))  
)
```



Histogram of Day3_Acc





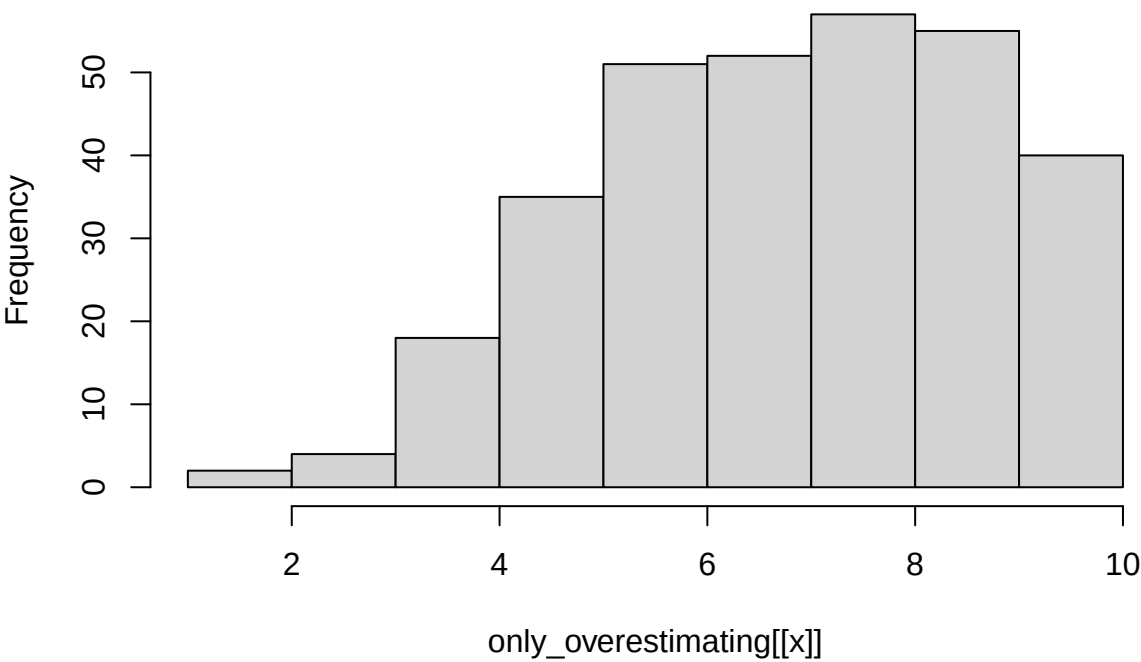
```
## [[1]]
## $breaks
## [1] 0 1 2 3 4 5 6 7 8 9
##
## $counts
## [1] 34 64 103 52 38 9 4 7 3
##
## $density
## [1] 0.10828025 0.20382166 0.32802548 0.16560510 0.12101911 0.02866242 0.01273885
## [8] 0.02229299 0.00955414
##
## $mids
## [1] 0.5 1.5 2.5 3.5 4.5 5.5 6.5 7.5 8.5
##
## $xname
## [1] "only_overestimating[[x]]"
##
## $equidist
## [1] TRUE
##
## attr(,"class")
## [1] "histogram"
##
## [[2]]
## $breaks
## [1] 0 1 2 3 4 5 6 7 8 9 10
```

```
##
## $counts
## [1] 60 23 43 37 37 19 18 18 27 32
##
## $density
## [1] 0.19108280 0.07324841 0.13694268 0.11783439 0.11783439 0.06050955
## [7] 0.05732484 0.05732484 0.08598726 0.10191083
##
## $mids
## [1] 0.5 1.5 2.5 3.5 4.5 5.5 6.5 7.5 8.5 9.5
##
## $xname
## [1] "only_overestimating[[x]]"
##
## $equidist
## [1] TRUE
##
## attr("class")
## [1] "histogram"
##
## [[3]]
## $breaks
## [1] -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8
##
## $counts
## [1] 9 11 25 24 58 48 37 27 27 17 13 16 2
##
## $density
## [1] 0.028662420 0.035031847 0.079617834 0.076433121 0.184713376 0.152866242
## [7] 0.117834395 0.085987261 0.085987261 0.054140127 0.041401274 0.050955414
## [13] 0.006369427
##
## $mids
## [1] -4.5 -3.5 -2.5 -1.5 -0.5 0.5 1.5 2.5 3.5 4.5 5.5 6.5 7.5
##
## $xname
## [1] "only_overestimating[[x]]"
##
## $equidist
## [1] TRUE
##
## attr("class")
## [1] "histogram"
```

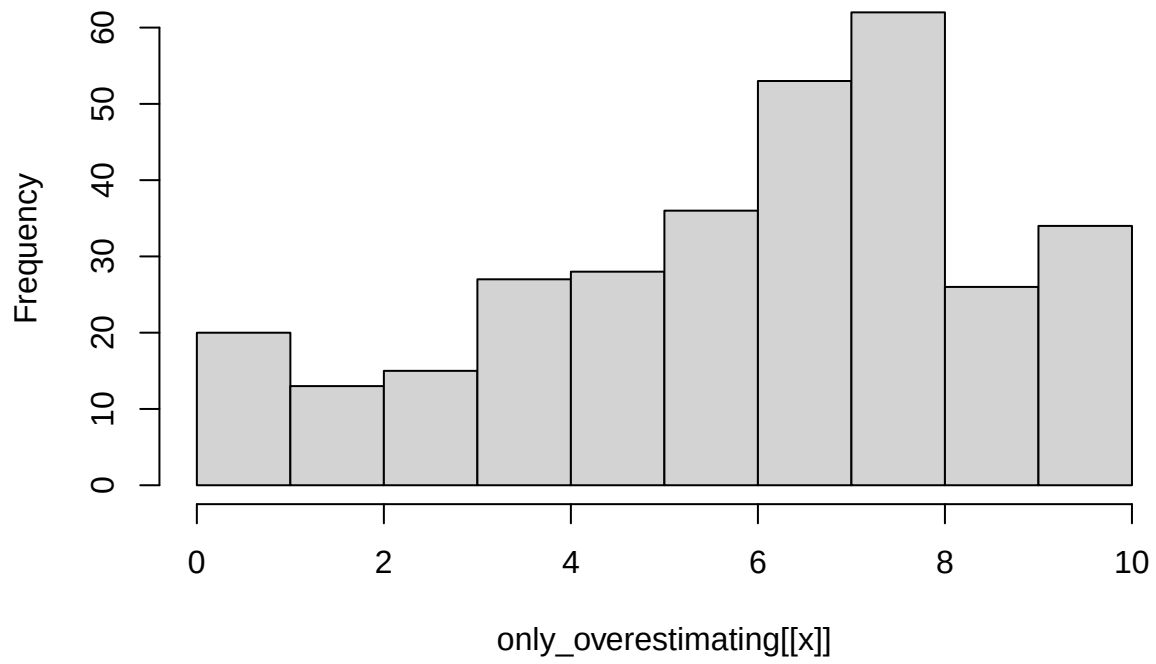
All postdictions

```
lapply(
  c("Day1_Postdiction", "Day3_Postdiction", "Postdiction_Change"),
  function(x)
    hist(only_overestimating[[x]], main = paste("Histogram of", x))
)
```

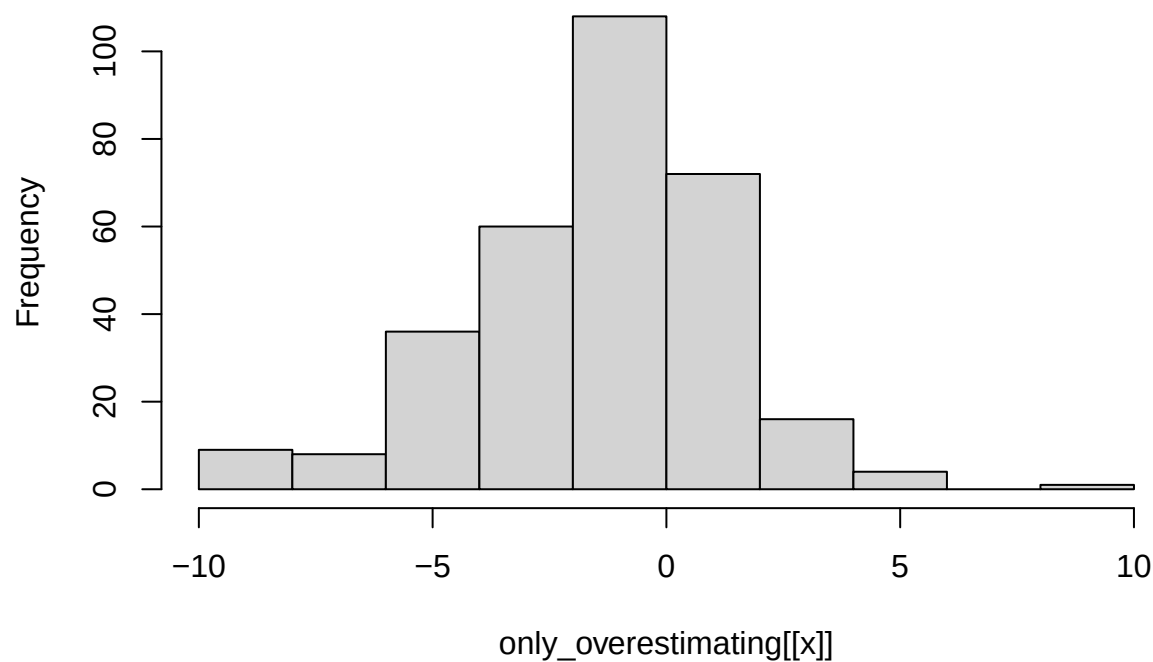

Histogram of Day1_Postdiction



Histogram of Day3_Postdiction



Histogram of Postdiction_Change



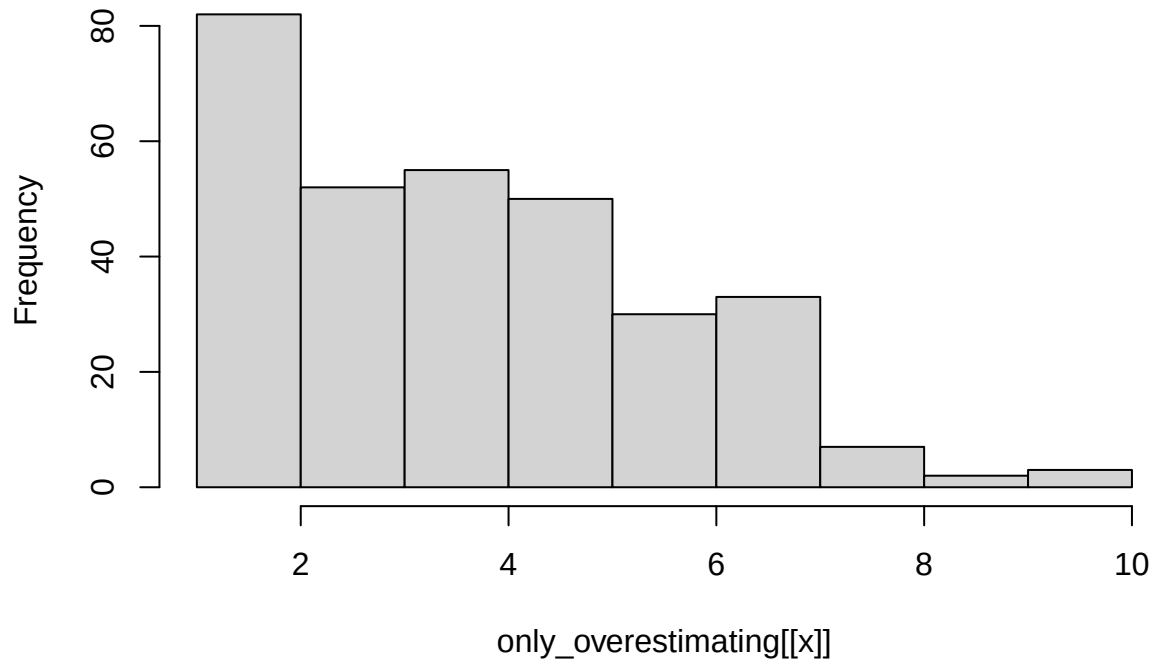
```
## [[1]]
## $breaks
## [1] 1 2 3 4 5 6 7 8 9 10
##
## $counts
## [1] 2 4 18 35 51 52 57 55 40
##
## $density
## [1] 0.006369427 0.012738854 0.057324841 0.111464968 0.162420382 0.165605096
## [7] 0.181528662 0.175159236 0.127388535
##
## $mids
## [1] 1.5 2.5 3.5 4.5 5.5 6.5 7.5 8.5 9.5
##
## $xname
## [1] "only_overestimating[[x]]"
##
## $equidist
## [1] TRUE
##
## attr(,"class")
## [1] "histogram"
##
## [[2]]
## $breaks
## [1] 0 1 2 3 4 5 6 7 8 9 10
```

```
##
## $counts
## [1] 20 13 15 27 28 36 53 62 26 34
##
## $density
## [1] 0.06369427 0.04140127 0.04777070 0.08598726 0.08917197 0.11464968
## [7] 0.16878981 0.19745223 0.08280255 0.10828025
##
## $mids
## [1] 0.5 1.5 2.5 3.5 4.5 5.5 6.5 7.5 8.5 9.5
##
## $xname
## [1] "only_overestimating[[x]]"
##
## $equidist
## [1] TRUE
##
## attr("class")
## [1] "histogram"
##
## [[3]]
## $breaks
## [1] -10 -8 -6 -4 -2 0 2 4 6 8 10
##
## $counts
## [1] 9 8 36 60 108 72 16 4 0 1
##
## $density
## [1] 0.014331210 0.012738854 0.057324841 0.095541401 0.171974522 0.114649682
## [7] 0.025477707 0.006369427 0.000000000 0.001592357
##
## $mids
## [1] -9 -7 -5 -3 -1 1 3 5 7 9
##
## $xname
## [1] "only_overestimating[[x]]"
##
## $equidist
## [1] TRUE
##
## attr("class")
## [1] "histogram"
```

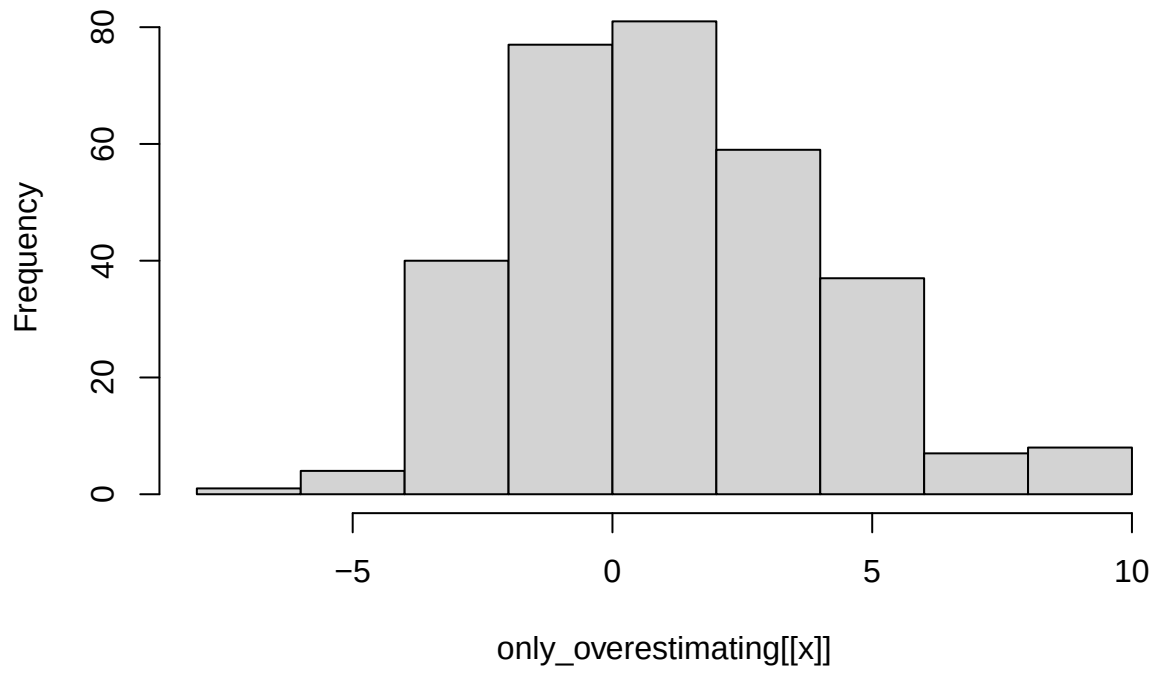
All metacognitions

```
lapply(
  c("Day1_Metacognition", "Day3_Metacognition", "Meta_Change"),
  function(x)
    hist(only_overestimating[[x]], main = paste("Histogram of", x))
)
```

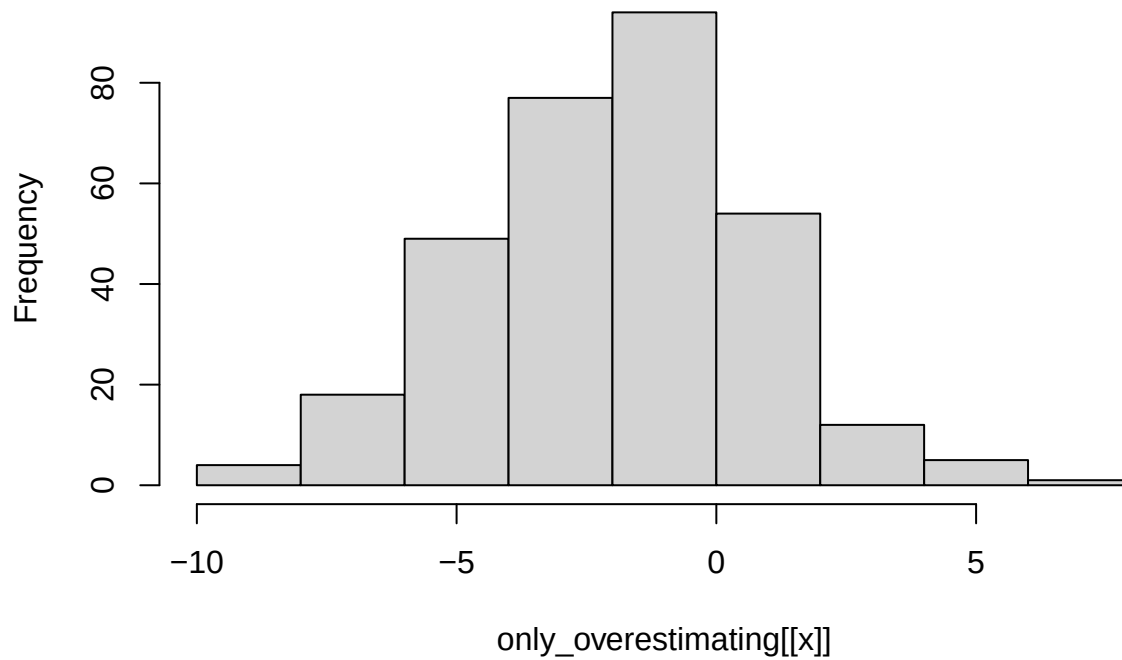
Histogram of Day1_Metacognition



Histogram of Day3_Metacognition



Histogram of Meta_Change



```
## [[1]]
## $breaks
## [1] 1 2 3 4 5 6 7 8 9 10
##
## $counts
## [1] 82 52 55 50 30 33 7 2 3
##
## $density
## [1] 0.261146497 0.165605096 0.175159236 0.159235669 0.095541401 0.105095541
## [7] 0.022292994 0.006369427 0.009554140
##
## $mids
## [1] 1.5 2.5 3.5 4.5 5.5 6.5 7.5 8.5 9.5
##
## $xname
## [1] "only_overestimating[[x]]"
##
## $equidist
## [1] TRUE
##
## attr(,"class")
## [1] "histogram"
##
## [[2]]
## $breaks
## [1] -8 -6 -4 -2 0 2 4 6 8 10
```

```

##
## $counts
## [1] 1 4 40 77 81 59 37 7 8
##
## $density
## [1] 0.001592357 0.006369427 0.063694268 0.122611465 0.128980892 0.093949045
## [7] 0.058917197 0.011146497 0.012738854
##
## $mids
## [1] -7 -5 -3 -1 1 3 5 7 9
##
## $xname
## [1] "only_overestimating[[x]]"
##
## $equidist
## [1] TRUE
##
## attr("class")
## [1] "histogram"
##
## [[3]]
## $breaks
## [1] -10 -8 -6 -4 -2 0 2 4 6 8
##
## $counts
## [1] 4 18 49 77 94 54 12 5 1
##
## $density
## [1] 0.006369427 0.028662420 0.078025478 0.122611465 0.149681529 0.085987261
## [7] 0.019108280 0.007961783 0.001592357
##
## $mids
## [1] -9 -7 -5 -3 -1 1 3 5 7
##
## $xname
## [1] "only_overestimating[[x]]"
##
## $equidist
## [1] TRUE
##
## attr("class")
## [1] "histogram"

```